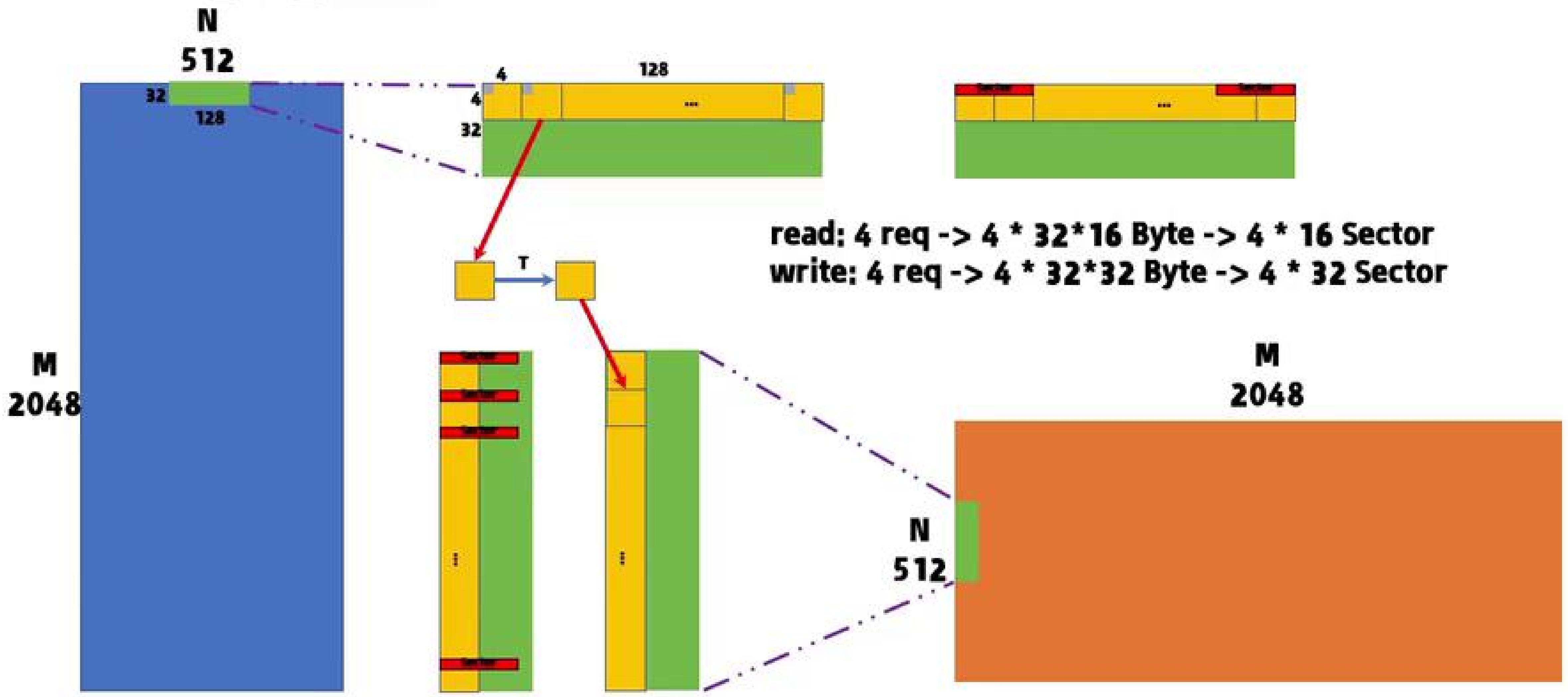
ncu与 kernel 调优指南 Occupancy Calculator

NVIDIA Nsight™ Compute

Transpose_v2_float4



read: 4 req -> 4 * 32*16 Byte -> 4 * 16 Sector
write: 4 req -> 4 * 32*32 Byte -> 4 * 32 Sector

block_size(32,8)

cuda性能优化笔记--Transpose性能优化 - 知乎

Ada GPU Architecture In-Depth

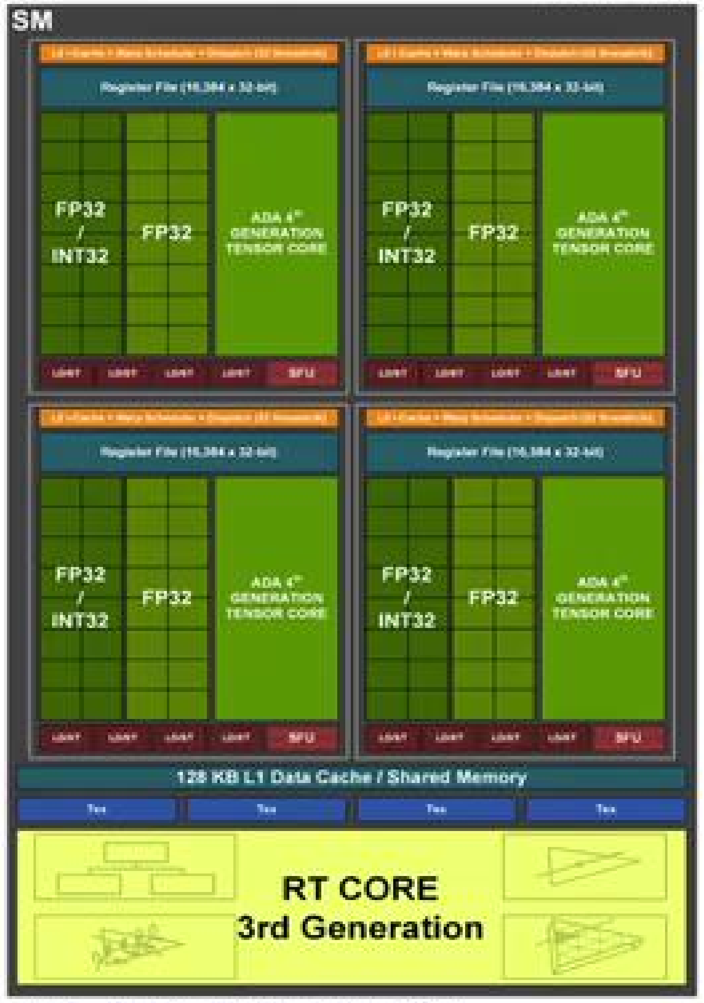


Figure 5. Ada Streaming Multiprocessor (SM)

occupancy初步理解

- occupancy中文翻译是占用率，一个SM中理论假如最大支持M个active warp（这里不理解什么是active warp可以参考这里），而实际上因为各种原因只能支持N个，这里N/M就是occupancy。

Active Warp（活跃线程束）：

- 在任何给定的时间点，GPU的SM（流处理器簇，Streaming Multiprocessors）可以同时处理多个warp。但是，由于资源限制（如执行单元、寄存器、共享内存等），并不是所有的warp都能同时活跃。
- Active warp指的是那些正在执行指令的warp。它们可能处于不同的执行阶段，比如正在执行计算任务、正在等待内存访问结果或正在执行控制流指令。

CUDA Occupancy Calculator占用率计算与显卡算力计算_cuda计算能力7.5-CSDN博客
GPU基础：Occupancy、wave and tail effect - 知乎
CUDA中Occupancy相关知识_cuda occupancy-CSDN博客

block_size对SM可启动warp数量的影响

Physical Limit of GPU (8.9):

Property	Limit
Threads per Warp	32
Max Warps per Multiprocessor	48
Max Thread Blocks per Multiprocessor	24
Max Threads per Multiprocessor	1536
Maximum Thread Block Size	1024
Registers per Multiprocessor	65536
Max Registers per Thread Block	65536
Max Registers per Thread	255
Shared Memory per Multiprocessor (bytes)	16384
Max Shared Memory per Block	16384
Register Allocation Unit Size	256
Register Allocation Granularity	warp
Shared Memory Allocation Unit Size	128
Warp Allocation Granularity	4
Shared Memory Per Block (bytes) (CUDA runtime use)	1024

1. 每一个Block必须分配到一个SM中
2. 每一个SM最多可放48个warp(1536个thread)
3. 每一个SM最多可放24个Block

假设需要计算的数据足够多，每个thread都只使用很少的smem和register，设置的Block_Size为(32,5)

- 根据2可得最多能放48个warp相当于9.6个Block
- 根据1可得0.6个Block是不能被分配到一个SM上的，所以只能放9个Block，相当于45个warp，所以occupany为0.9375=48/45

Allocated Resources:

Resources	Per Block	Limit Per SM	Allocatable Blocks Per SM
Warps (Threads Per Block / Threads Per Warp)	5	48	9
Registers (Warp limit per SM due to per-warp reg count)	5	256	51
Shared Memory (Bytes)	1024	65536	64

CUDA微架构与指令集（4）-指令发射与warp调度 - 知乎

warp scheduler与Achieved Occupancy（实际占用率）

SM详解与Warp Scheduler，合理块和线程的数量对GPU利用率非常重要 - 知乎

cuda的warp scheduler知识-CSDN博客

CUDA中Occupancy相关知识_cuda occupancy-CSDN博客

按照我个人的4090显卡来说，每个SM最多可以处理48个warp，而从架构图上4090上面有4个warp scheduler，所以每个warp scheduler可以负责issue 12个warp（有12个warp slots）

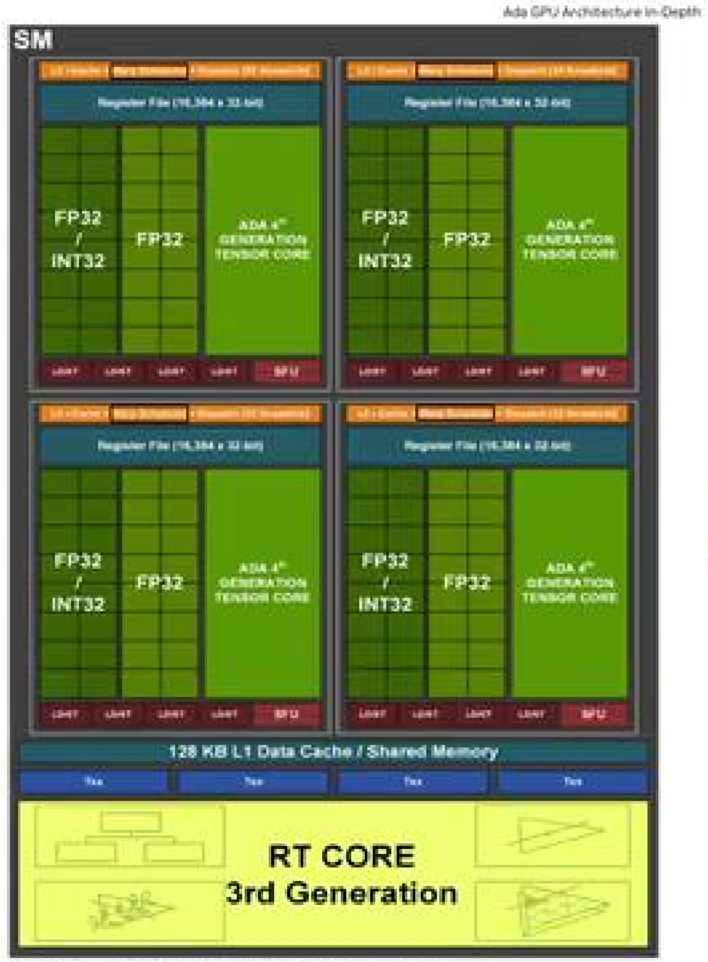
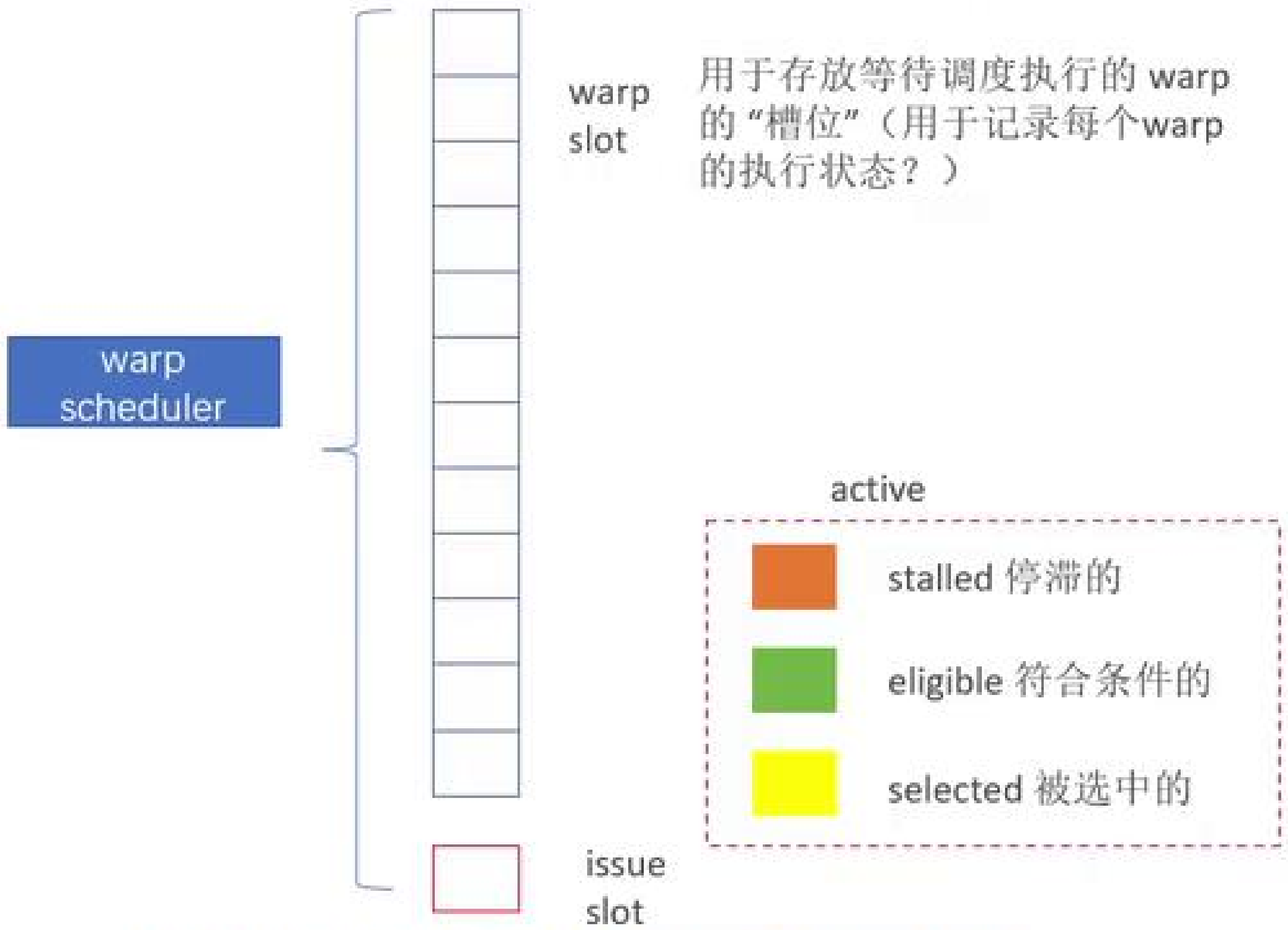


Figure 5. Ada Streaming Multiprocessor (SM)

Ada AD102 GPU SM



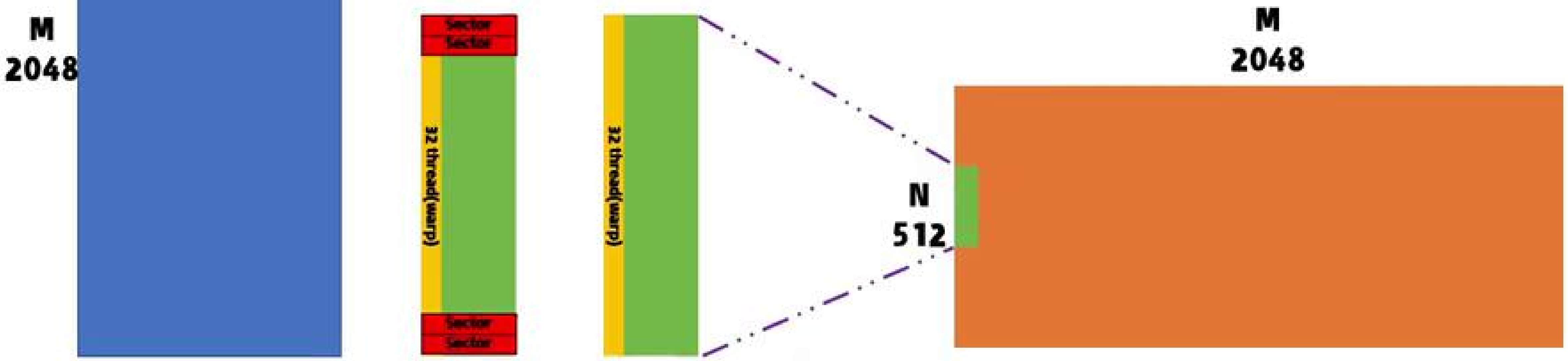
[NVIDIA GPU 架构与 CUDA 算力 - CuiYuhao's Blog]

[cuda的warp scheduler知识-CSDN博客]

[Nsight Compute(NCU) Scheduler Statistics 数据解读_nsys compute ncu-CSDN博客]

Transpose_v1_naive

read: 1req -> 4 Sector -> 4*32Byte
write: 1req -> 32 Sector -> 32*32 Byte



block_size(32,8)

cuda性能优化笔记--Transpose性能优化 - 知乎

Ada GPU Architecture In-Depth

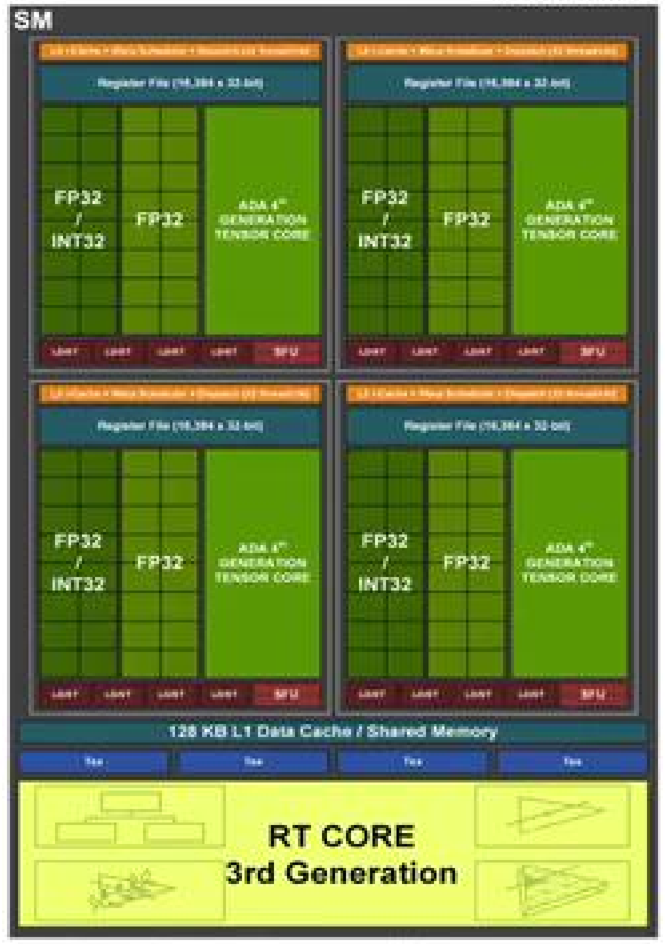


Figure 5. Ada Streaming Multiprocessor (SM)

以左侧图为例

可看出一个SM中有128个Core，所以至少是可以放下128个Thread（4个warp）为了隐藏延时，其实可以放进去更多warp（大于4个）根据显卡的架构不同，这个更多的warp仍然有一个理论上限。

1.用户设置的block的大小会影响实际SM上可以容纳的warp数量

2.CUDA隐藏延迟的手段就是当一个warp处于等待状态的时候会立刻切到下一个warp，而这个立刻就要求线程的切换要特别的迅速（0开销），这与CPU多线程有很大的区别，CPU在进行thread切换的时候，是有开销的（要保留每个thread的上下文），但是GPU不需要因为它有大量的寄存器，每个thread的寄存器数据都是自己用所以就没有切换上下文的需要，自然也可以做到0开销，而与此同时每个SM中的寄存器数量是有上限的所以寄存器也会成为能开启多少个warp的一个衡量指标

3.共享内存和寄存器类似，每个SM只有固定大小，所以也影响warp（block块）的数量。

123再结合木桶效应，取最小值就可以得到每个SM能容纳的warp数量。

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123再结合木桶效应，取最小值就可以得到每个SM能容纳的warp数量。

CUDA Occupancy Calculator

Compute Capability:	8.9	Threads Per Block:	32
Shared Memory Size Config (bytes):	32768	Registers Per Thread:	1
Global Load Cache Mode:	L1+L2 (ca)	User Shared Memory Per Block (bytes):	0

setting

input

✓ Apply Automatically   Apply   Reset

Tables   Utilization   Graphs   GPU Data

Occupancy Data:

Property		Value
Active Threads per Multiprocessor	每一个SM中活跃的Threads	768
Active Warps per Multiprocessor	每一个SM中活跃的Warps	24
Active Thread Blocks per Multiprocessor	每一个SM中活跃的Blocks	24
Occupancy of each Multiprocessor	Occupancy	50 %

Allocated Resources:

Resources	Per Block	Limit Per SM	Allocatable Blocks Per SM
Warps (Threads Per Block / Threads Per Warp)	1	48	24
Registers (Warp limit per SM due to per-warp reg count)	1	256	256
Shared Memory (Bytes)	1024	32768	32

Occupancy Limiters:

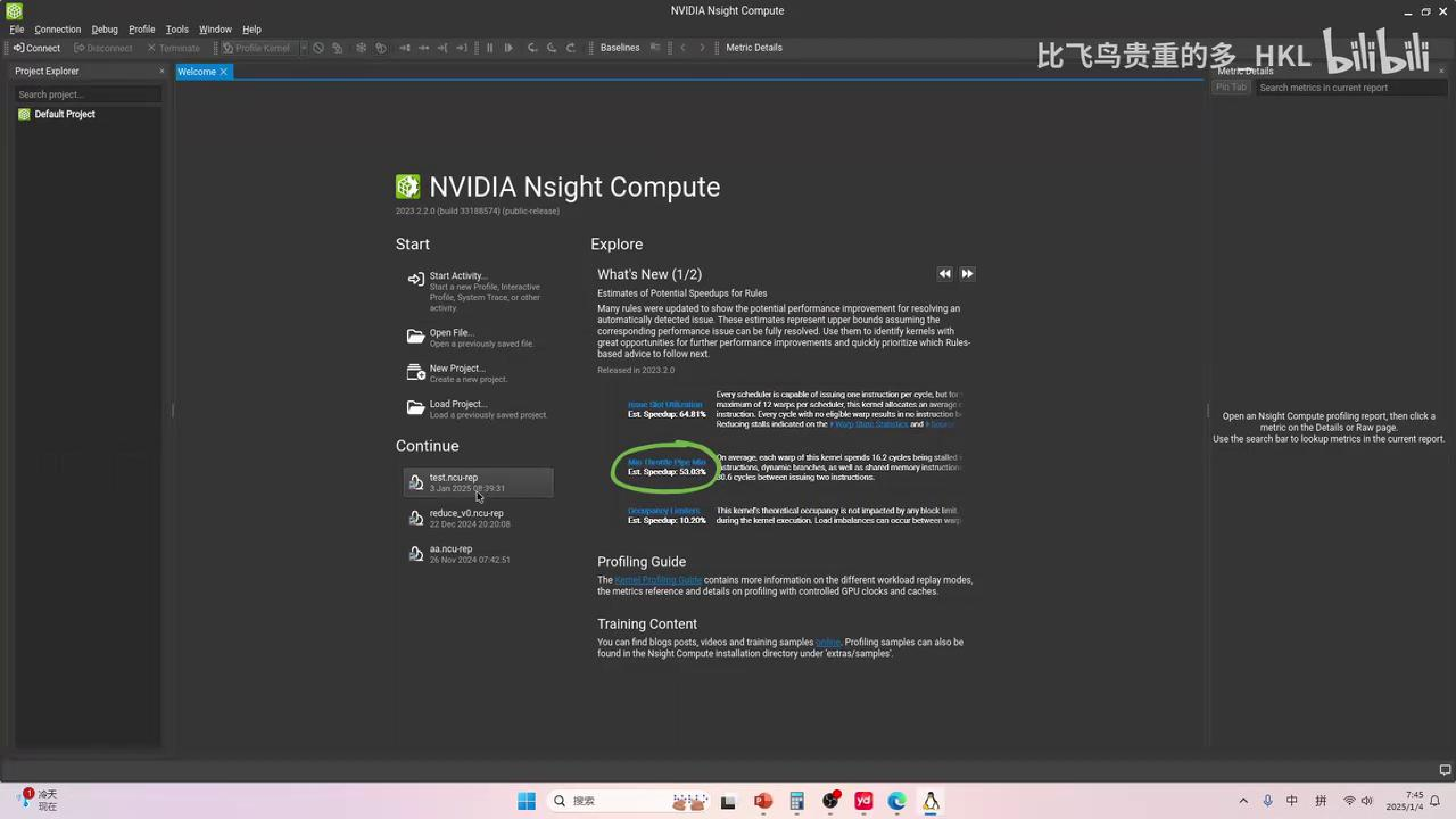
Limited By	Blocks per SM	Warps Per Block	Warps Per SM
Max Warps or Max Blocks per Multiprocessor	24	1 ⚠	24
Registers per Multiprocessor	256		
Shared Memory per Multiprocessor	32		

Physical Limit of GPU (8.9):

Property	Limit
Threads per Warp	32
Max Warps per Multiprocessor	48
Max Thread Blocks per Multiprocessor	24
Max Threads per Multiprocessor	1536
Maximum Thread Block Size	1024
Registers per Multiprocessor	65536
Max Registers per Thread Block	65536
Max Registers per Thread	255
Shared Memory per Multiprocessor (bytes)	32768
Max Shared Memory per Block	32768
Register Allocation Unit Size	256
Register Allocation Granularity	warp
Shared Memory Allocation Unit Size	128
Warp Allocation Granularity	4
Shared Memory Per Block (bytes) (CUDA runtime use)	1024

Physical Limit of GPU

CUDA Occupancy Calculator


NVIDIA Nsight Compute
File
Connection
Debug
Profile
Tools
Window
Help
Connect
Disconnect
Terminate
Profile Kernel
Baselines
Metric Details
比飞鸟贵重的多_HKL
bilibili
Project Explorer
Search project...
Default Project
Welcome
Metric Details
Pin Tab
Search metrics in current report
NVIDIA Nsight Compute
2023.2.2.0 (build 33188574) (public-release)
Start
Start Activity...
Start a new Profile, Interactive Profile, System Trace, or other activity.
Open File...
Open a previously saved file.
New Project...
Create a new project.
Load Project...
Load a previously saved project.
Continue
test.ncu-rep
3 Jan 2025 08:39:31
reduce_v0.ncu-rep
22 Dec 2024 20:20:08
aa.ncu-rep
26 Nov 2024 07:42:51
Explore
What's New (1/2)
Estimates of Potential Speedups for Rules
Many rules were updated to show the potential performance improvement for resolving an automatically detected issue. These estimates represent upper bounds assuming the corresponding performance issue can be fully resolved. Use them to identify kernels with great opportunities for further performance improvements and quickly prioritize which Rules-based advice to follow next.
Released in 2023.2.0
Est. Speedup: 64.81%
Est. Speedup: 53.03%
Est. Speedup: 10.20%
Profiling Guide
The Kernel Profiling Guide contains more information on the different workload replay modes, the metrics reference and details on profiling with controlled GPU clocks and caches.
Training Content
You can find blogs posts, videos and training samples online. Profiling samples can also be found in the Nsight Compute installation directory under 'extras/samples'.
Open an Nsight Compute profiling report, then click a metric on the Details or Raw page.
Use the search bar to lookup metrics in the current report.
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NVIDIA Nsight Compute

File Connection Debug Profile Tools Window Help

Connect Disconnect Terminate Profile Kernel Baselines Metric Details

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Project Explorer

Search project...

Default Project

test.ncu-rep

Welcome test.ncu-rep

Page: Summary Result: 0 - 529 - transpose_naive Add Baseline Apply Rules Occupancy Calculator Copy as Image

	Result	Time	Cycles	Regs	GPU	SM Frequency	CC	Process
Current	529 - transpose_naive (64, 64, 1)x(8, 32, 1)	9.57 usecond	21,596	16	0 - NVIDIA GeForce RTX 4090	2.25 cycle/nsecond	8.9	[544] my_transpose

Use the column headers to sort the results in this report. Double-click a result to see detailed metrics.

ID	Estimated Speedup	Function Name	Demangled Name	Duration	Runtime Improvement	Compute Throughput	Memory Throughput	# Registers	Grid Size
0	92.97	transpose_naive	transpose_naive(flo...	9.57	8.90	9.50	71.00	16	64,
1	92.93	transpose_naive	transpose_naive(flo...	8.90	8.27	10.73	75.57	16	64,
2	92.97	transpose_naive	transpose_naive(flo...	8.96	8.33	10.79	75.77	16	64,
3	92.88	transpose_naive	transpose_naive(flo...	8.80	8.17	10.51	74.54	16	64,
4	92.87	transpose_naive	transpose_naive(flo...	8.96	8.32	10.71	75.16	16	64,
5	92.96	transpose_naive	transpose_naive(flo...	8.96	8.33	10.64	75.62	16	64,
6	92.93	transpose_naive	transpose_naive(flo...	8.64	8.03	10.73	75.37	16	64,
7	92.93	transpose_naive	transpose_naive(flo...	8.93	8.30	8.66	75.44	16	64,
8	92.97	transpose_naive	transpose_naive(flo...	8.86	8.24	10.07	70.73	16	64,
9	92.99	transpose_naive	transpose_naive(flo...	11.07	10.30	10.63	84.05	16	64,
10	97.01	transpose_float4_in...	void transpose_floa...	7.30	7.08	2.38	82.47	38	16,
11	96.97	transpose_float4_in...	void transpose_floa...	7.23	7.01	2.80	71.00	38	16,
12	96.95	transpose_float4_in...	void transpose_floa...	7.04	6.82	2.82	71.31	38	16,
13	96.94	transpose_float4_in...	void transpose_floa...	7.23	7.01	1.97	82.94	38	16,
14	97.00	transpose_float4_in...	void transpose_floa...	8.29	8.04	2.77	72.35	38	16,
15	96.95	transpose_float4_in...	void transpose_floa...	8.51	8.25	2.88	60.56	38	16,

The following performance optimization opportunities were discovered for this result. Follow the rule links to see more context on the Details page.
Note: *Speedup estimations are experimental and might overestimate optimization potential.*

Issue Slot Utilization
Est. Speedup: 92.97%

Every scheduler is capable of issuing one instruction per cycle, but for this kernel each scheduler only issues an instruction every 14.2 cycles. This might leave hardware resources underutilized and may lead to less optimal performance. Out of the maximum of 12 warps per scheduler, this kernel allocates an average of 8.70 active warps per scheduler, but only an average of 0.11 warps were eligible per cycle. Eligible warps are the subset of active warps that are ready to issue their next instruction. Every cycle with no eligible warp results in no instruction being issued and the issue slot remains unused. To increase the number of eligible warps, reduce the time the active warps are stalled by inspecting the top stall reasons on the Warp State Statistics and Source Counters sections.

Long Scoreboard
Est. Speedup: 84.50%

On average, each warp of this kernel spends 104.6 cycles being stalled waiting for a scoreboard dependency on a L1TEX (local, global, surface, texture) operation. Find the instruction producing the data being waited upon to identify the culprit. To reduce the number of cycles waiting on L1TEX data accesses verify the memory access patterns are optimal for the target architecture, attempt to increase cache hit rates by increasing data locality (coalescing), or by changing the cache configuration. Consider moving frequently used data to shared memory. This stall type represents about 84.5% of the total average of 123.8 cycles between issuing two instructions.

Occupancy Limiters
Est. Speedup: 27.40%

This kernel's theoretical occupancy is not impacted by any block limit. The difference between calculated theoretical (100.0%) and measured achieved occupancy (72.6%) can be the result of warp scheduling overheads or workload imbalances during the kernel execution. Load imbalances can occur between warps within a block as well as across blocks of the same kernel. See the CUDA Best Practices Guide for more details on optimizing occupancy.

Metric Details

Pin Tab Search metrics in current report

Open an Nsight Compute profiling report, then click a metric on the Details or Raw page.
Use the search bar to lookup metrics in the current report.

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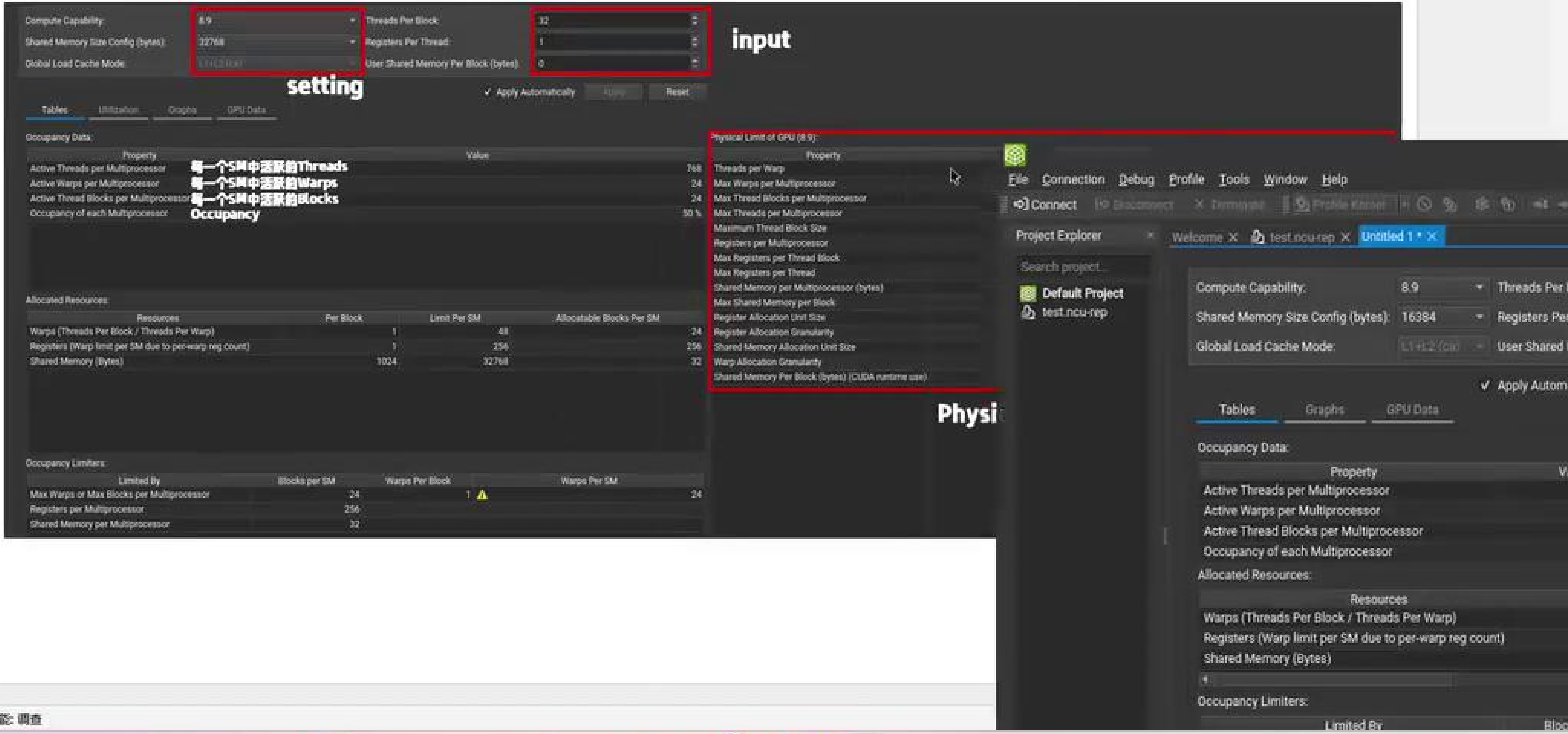
中 拼 7:45 2025/1/4

CUDA Occupancy Calculator



CUDA Occupancy Calculator

NVIDIA Nsight Compute

File Connection Debug Profile Tools Window Help

Project Explorer

Default Project
test.ncu-rep

Welcome | test.ncu-rep | Untitled 1 *

Compute Capability:	8.9	Threads Per Block:	256
Shared Memory Size Config (bytes):	16384	Registers Per Thread:	16
Global Load Cache Mode:	L1+L2 (ca)	User Shared Memory Per Block (bytes):	0

✓ Apply Automatically Apply Reset

Tables | Graphs | GPU Data

Occupancy Data:

Property	Value
Active Threads per Multiprocessor	1536
Active Warps per Multiprocessor	48
Active Thread Blocks per Multiprocessor	6
Occupancy of each Multiprocessor	100 %

Allocated Resources:

Resources	Per Block
Warps (Threads Per Block / Threads Per Warp)	
Registers (Warp limit per SM due to per-warp reg count)	
Shared Memory (Bytes)	

Occupancy Limiters:

Limited By	Blocks per SM
Max Warps or Max Blocks per Multiprocessor	6
Registers per Multiprocessor	16
Shared Memory per Multiprocessor	16

Physical Limit of GPU (8.9):

Property	Li
Threads per Warp	
Max Warps per Multiprocessor	
Max Thread Blocks per Multiprocessor	
Max Threads per Multiprocessor	
Maximum Thread Block Size	
Registers per Multiprocessor	
Max Registers per Thread Block	
Max Registers per Thread	
Shared Memory per Multiprocessor (bytes)	
Max Shared Memory per Block	
Register Allocation Unit Size	
Register Allocation Granularity	
Shared Memory Allocation Unit Size	
Warp Allocation Granularity	
Shared Memory Per Block (bytes) (CUDA runtime use)	

Metric Details

Open an Nsight Compute profiling report, then click a metric on the Details or Raw page.
Use the search bar to lookup metrics in the current report.

CUDA Occupancy Calculator

Compute Capability: 8.9
Shared Memory Size Config (bytes): 32768
Global Load Cache Mode:

setting

Threads Per Block: 32
Registers Per Thread: 1
User Shared Memory Per Block (bytes): 0

input

✓ Apply Automatically Apply Reset

Tables Utilization Graphs GPU Data

Occupancy Data:

Property		Value
Active Threads per Multiprocessor	每一个SM中活跃的Threads	768
Active Warps per Multiprocessor	每一个SM中活跃的Warps	24
Active Thread Blocks per Multiprocessor	每一个SM中活跃的Blocks	24
Occupancy of each Multiprocessor	Occupancy	50 %

Allocated Resources:

Resources	Per Block	Limit Per SM	Allocatable Blocks Per SM
Warps (Threads Per Block / Threads Per Warp)	1	48	24
Registers (Warp limit per SM due to per-warp reg count)	1	256	256
Shared Memory (Bytes)	1024	32768	32

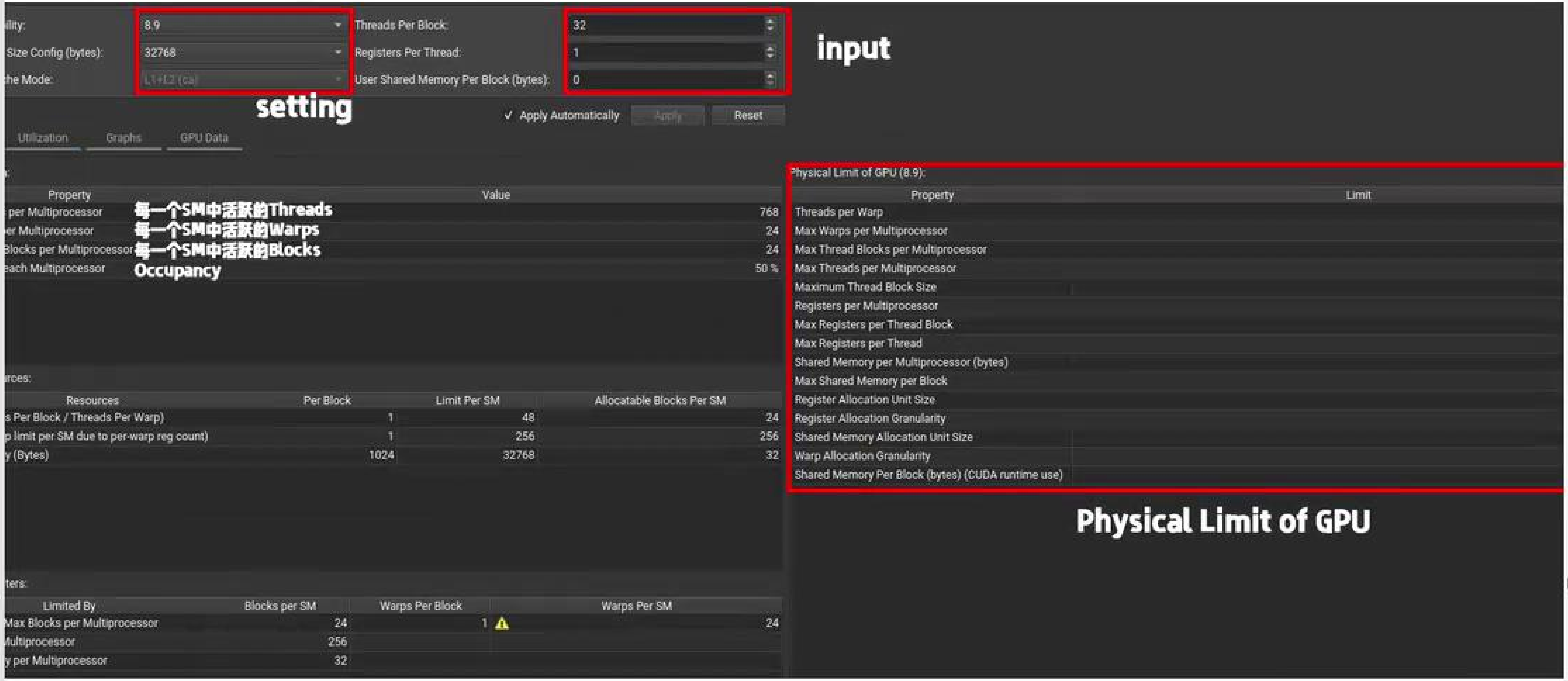
Occupancy Limiters:

Limited By	Blocks per SM	Warps Per Block	Warps Per SM
Max Warps or Max Blocks per Multiprocessor	24	1 ⚠	24
Registers per Multiprocessor	256		
Shared Memory per Multiprocessor	32		

Physical Limit of GPU (8.9)

Property	Limit
Threads per Warp	32
Max Warps per Multiprocessor	48
Max Thread Blocks per Multiprocessor	24
Max Threads per Multiprocessor	1536
Maximum Thread Block Size	1024
Registers per Multiprocessor	65536
Max Registers per Thread Block	65536
Max Registers per Thread	255
Shared Memory per Multiprocessor (bytes)	32768
Max Shared Memory per Block	32768
Register Allocation Unit Size	256
Register Allocation Granularity	warp
Shared Memory Allocation Unit Size	128
Warp Allocation Granularity	4
Shared Memory Per Block (bytes) (CUDA runtime use)	1024

Physical Limit of GPU


ility: 8.9
Size Config (bytes): 32768
che Mode: L1+L2 (ca)
Threads Per Block: 32
Registers Per Thread: 1
User Shared Memory Per Block (bytes): 0
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Utilization
Graphs
GPU Data
Property
Value
per Multiprocessor 每一个SM中活跃的Threads 768
er Multiprocessor 每一个SM中活跃的Warps 24
Blocks per Multiprocessor 每一个SM中活跃的Blocks 24
each Multiprocessor Occupancy 50 %
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Allocatable Blocks Per SM
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p limit per SM due to per-warp reg count) 1 256 256
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ters:
Limited By
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Warps Per Block
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Max Blocks per Multiprocessor 24 1 24
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Physical Limit of GPU (8.9):
Property
Limit
Threads per Warp
Max Warps per Multiprocessor
Max Thread Blocks per Multiprocessor
Max Threads per Multiprocessor
Maximum Thread Block Size
Registers per Multiprocessor
Max Registers per Thread Block
Max Registers per Thread
Shared Memory per Multiprocessor (bytes)
Max Shared Memory per Block
Register Allocation Unit Size
Register Allocation Granularity
Shared Memory Allocation Unit Size
Warp Allocation Granularity
Shared Memory Per Block (bytes) (CUDA runtime use)
Physical Limit of GPU

Compute Capability: 8.9 | Threads Per Block: 32

Shared Memory Size Config (bytes): 32768 | Registers Per Thread: 1

input

NVIDIA Nsight Compute

Compute Capability: 8.9

Threads Per Block: 256

Shared Memory Size Config (bytes): 16384

Registers Per Thread: 16

Global Load Cache Mode: L1+L2 (ca)

User Shared Memory Per Block (bytes): 0

✓ Apply Automatically | Apply | Reset

Tables | Graphs | GPU Data

Occupancy Data:

Property	Value
Active Threads per Multiprocessor	1536
Active Warps per Multiprocessor	48
Active Thread Blocks per Multiprocessor	6
Occupancy of each Multiprocessor	100 %

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Shared Memory (Bytes)	

Occupancy Limiters:

Limited By	Blocks per SM
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Physical Limit of GPU (8.9):

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Max Thread Blocks per Multiprocessor	
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Registers per Multiprocessor	
Max Registers per Thread Block	
Max Registers per Thread	
Shared Memory per Multiprocessor (bytes)	
Max Shared Memory per Block	
Register Allocation Unit Size	
Register Allocation Granularity	
Shared Memory Allocation Unit Size	
Warp Allocation Granularity	
Shared Memory Per Block (bytes) (CUDA runtime use)	

Metric Details

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cal Limit of GPU

Compute Capability:	8.9	Threads Per Block:	32
Shared Memory Size Config (bytes):	32768	Registers Per Thread:	1
Global Load Cache Mode:	L1+L2 (ca)	User Shared Memory Per Block (bytes):	0

input

setting

✓ Apply Automatically Apply Reset

Tables Utilization Graphs GPU Data

Occupancy Data:

Property		Value
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Occupancy of each Multiprocessor	Occupancy	50 %

Allocated Resources:

Resources	Per Block	Limit Per SM	Allocatable Blocks Per SM
Warps (Threads Per Block / Threads Per Warp)	1	48	24
Registers (Warp limit per SM due to per-warp reg count)	1	256	256
Shared Memory (Bytes)	1024	32768	32

Occupancy Limiters:

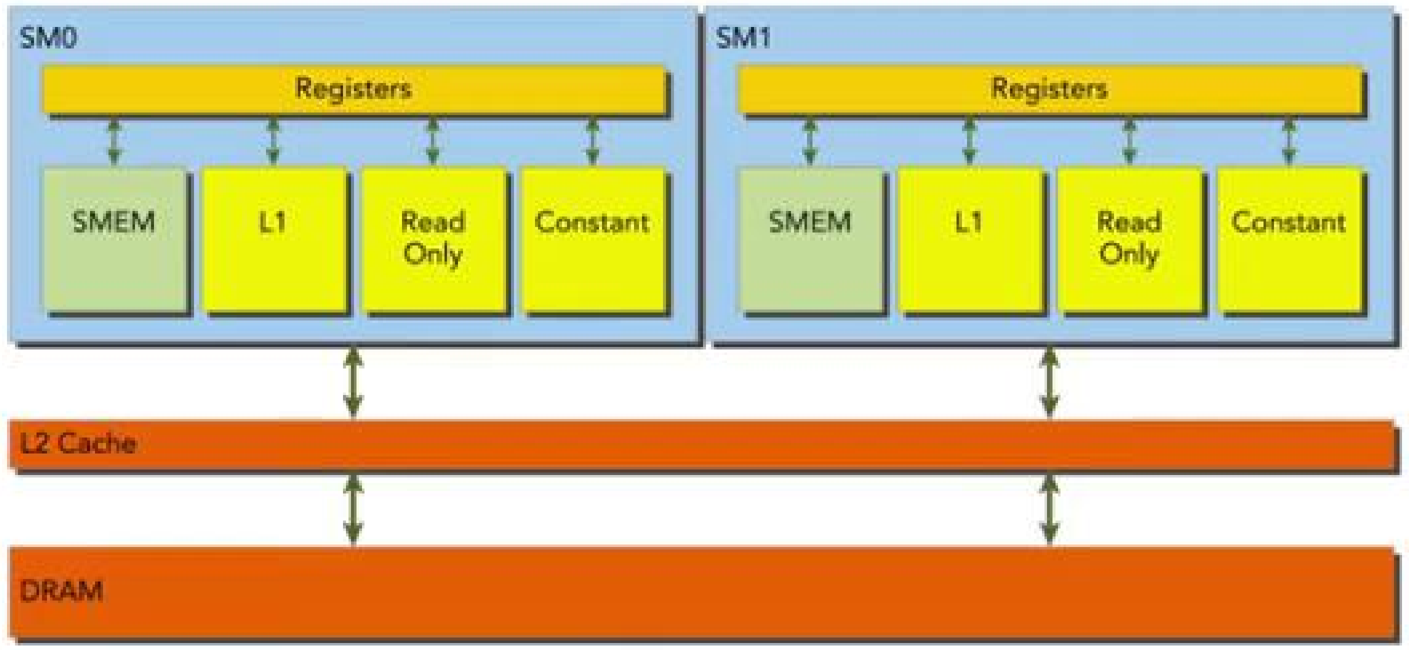
Limited By	Blocks per SM	Warps Per Block	Warps Per SM
Max Warps or Max Blocks per Multiprocessor	24	1 ⚠	24
Registers per Multiprocessor	256		
Shared Memory per Multiprocessor	32		

Physical Limit of GPU (8.9):

Property	Limit
Threads per Warp	
Max Warps per Multiprocessor	
Max Thread Blocks per Multiprocessor	
Max Threads per Multiprocessor	
Maximum Thread Block Size	
Registers per Multiprocessor	
Max Registers per Thread Block	
Max Registers per Thread	
Shared Memory per Multiprocessor (bytes)	
Max Shared Memory per Block	
Register Allocation Unit Size	
Register Allocation Granularity	
Shared Memory Allocation Unit Size	
Warp Allocation Granularity	
Shared Memory Per Block (bytes) (CUDA runtime use)	

Physical Limit of GPU

2.1 对齐合并访问



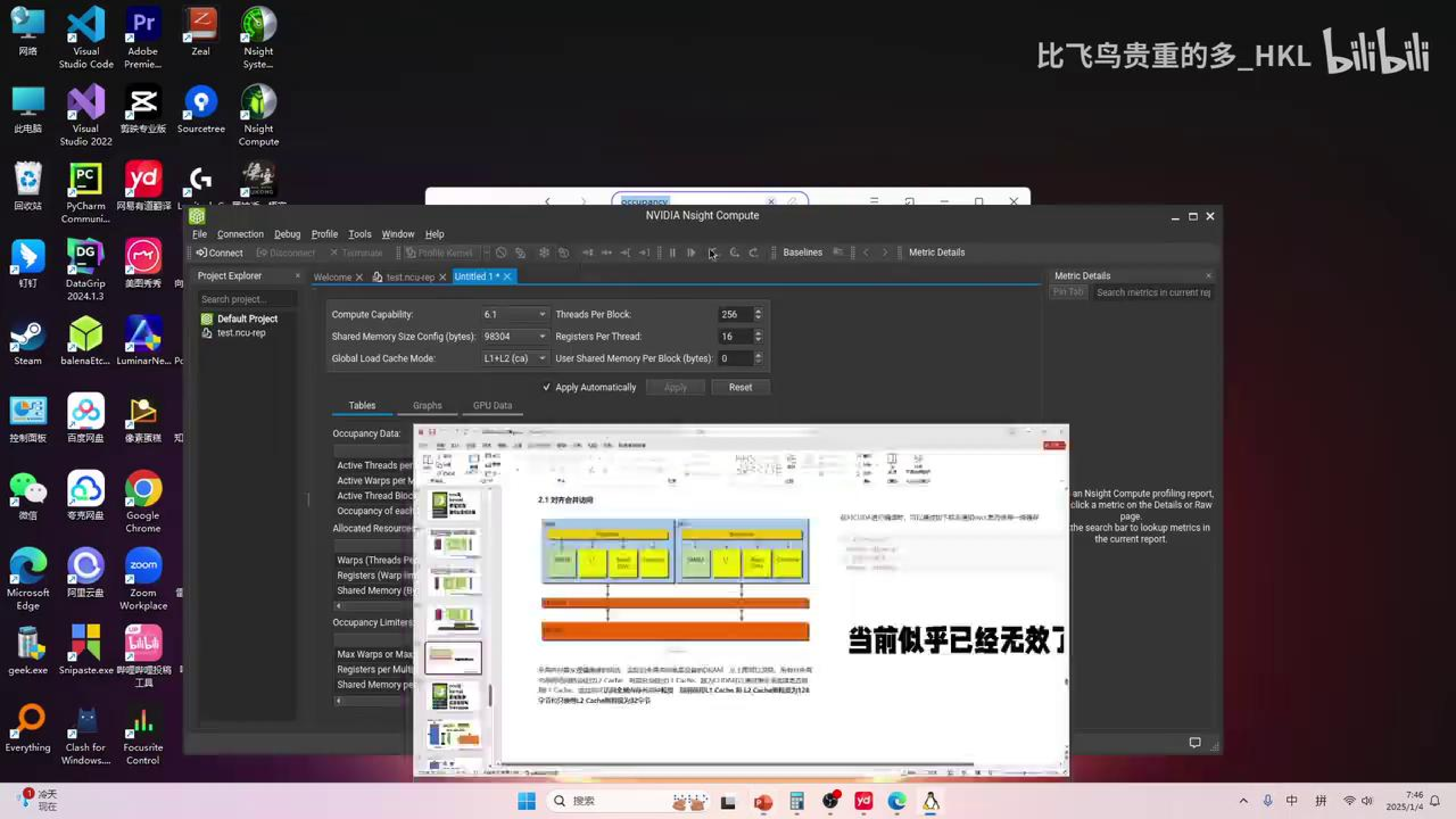
DRAM

全局内存是从逻辑角度的叫法，实际的全局内存就是设备的DRAM，从上图可以发现，所有对全局内存的访问都会经过L2 Cache，有部分会经过L1 Cache。因为CUDA可以通过指令来选择是否启用L1 Cache，由此每次**访问全局内存**有两种**粒度**：**同时使用L1 Cache 和 L2 Cache则粒度为128字节**和**只使用L2 Cache则粒度为32字节**

在对CUDA进行编译时，可以通过如下标志通知nvcc是否使用一级缓存

```
# 禁用一级缓存
-Xptxas -dlcm=cg
# 启用一级缓存
-Xptxas -dlcm=ca
```

当前似乎已经无效了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File Connection Debug Profile Tools Window Help
Connect Disconnect Terminate Profile Kernel Baselines Metric Details
Project Explorer
Search project...
Default Project
test.ncu-rep
Welcome test.ncu-rep Untitled 1 *
Compute Capability: 6.1
Threads Per Block: 256
Shared Memory Size Config (bytes): 98304
Registers Per Thread: 16
Global Load Cache Mode: L1+L2 (ca)
User Shared Memory Per Block (bytes): 0
Apply Automatically Apply Reset
Tables Graphs GPU Data
Occupancy Data:
Active Threads per
Active Warps per M
Active Thread Bloc
Occupancy of each
Allocated Resource
Warps (Threads Pe
Registers (Warp lim
Shared Memory (B
Occupancy Limiters
Max Warps or Max
Registers per Mult
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2.1 对齐合并访问
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Sourcetree
Nsight Compute
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PyCharm Communi...
网易有道翻译
钉钉
DataGrip 2024.1.3
美图秀秀
Steam
balenaEtc...
LuminarNe...
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百度网盘
微信
夸克网盘
Google Chrome
Microsoft Edge
阿里云盘
Zoom Workplace
geek.exe
Snipaste.exe
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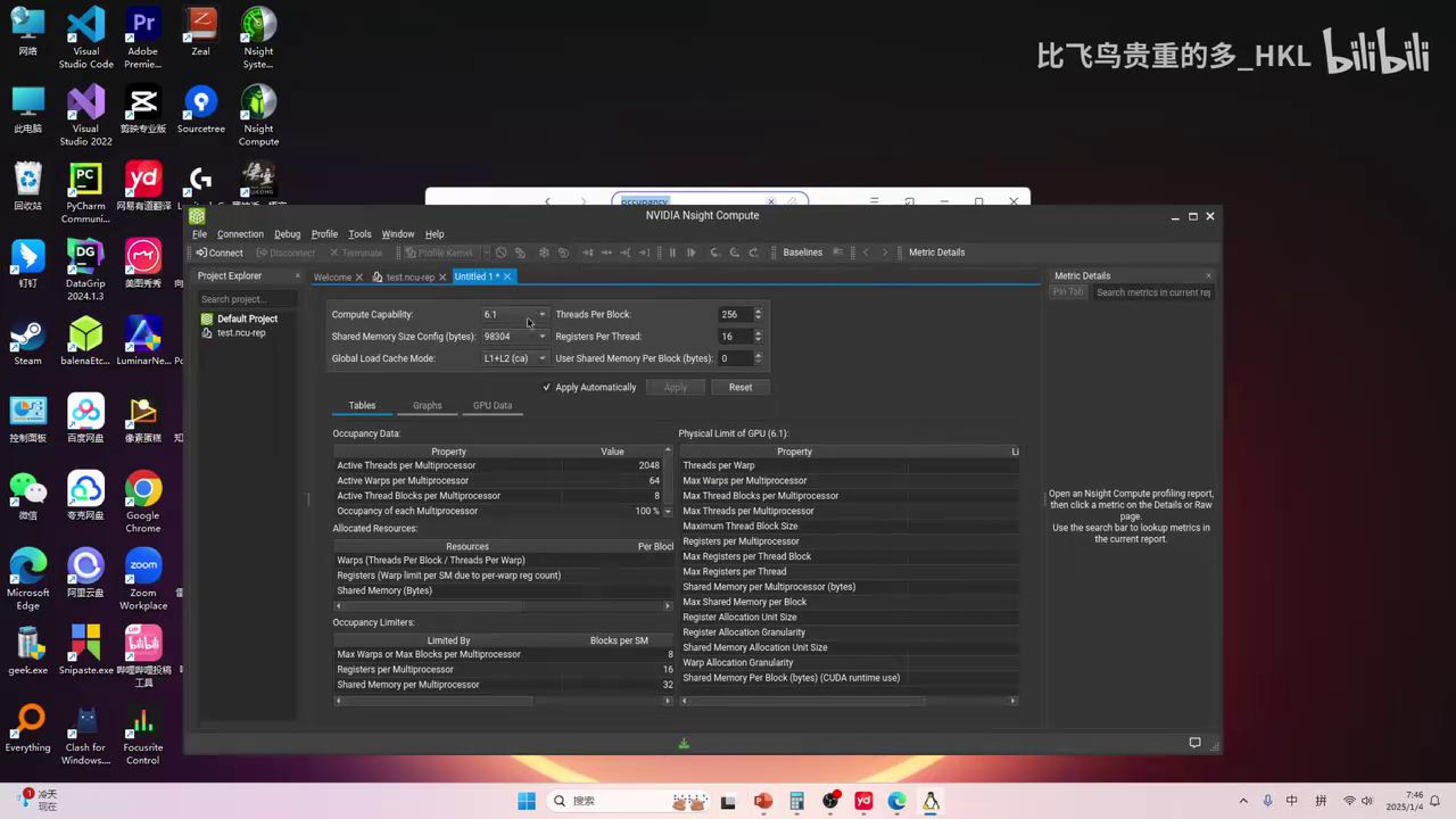

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Baselines
Metric Details
Project Explorer
Search project...
Default Project
test.ncu-rep
Welcome
test.ncu-rep
Untitled 1 *
Compute Capability: 6.1
Threads Per Block: 256
Shared Memory Size Config (bytes): 98304
Registers Per Thread: 16
Global Load Cache Mode: L1+L2 (ca)
User Shared Memory Per Block (bytes): 0
Apply Automatically
Apply
Reset
Tables
Graphs
GPU Data
Occupancy Data:
Property | Value
Active Threads per Multiprocessor | 2048
Active Warps per Multiprocessor | 64
Active Thread Blocks per Multiprocessor | 8
Occupancy of each Multiprocessor | 100 %
Allocated Resources:
Resources | Per Block
Warps (Threads Per Block / Threads Per Warp)
Registers (Warp limit per SM due to per-warp reg count)
Shared Memory (Bytes)
Occupancy Limiters:
Limited By | Blocks per SM
Max Warps or Max Blocks per Multiprocessor | 8
Registers per Multiprocessor | 16
Shared Memory per Multiprocessor | 32
Physical Limit of GPU (6.1):
Property
Threads per Warp
Max Warps per Multiprocessor
Max Thread Blocks per Multiprocessor
Max Threads per Multiprocessor
Maximum Thread Block Size
Registers per Multiprocessor
Max Registers per Thread Block
Max Registers per Thread
Shared Memory per Multiprocessor (bytes)
Max Shared Memory per Block
Register Allocation Unit Size
Register Allocation Granularity
Shared Memory Allocation Unit Size
Warp Allocation Granularity
Shared Memory Per Block (bytes) (CUDA runtime use)
Metric Details
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Search metrics in current rep
Open an Nsight Compute profiling report, then click a metric on the Details or Raw page.
Use the search bar to lookup metrics in the current report.
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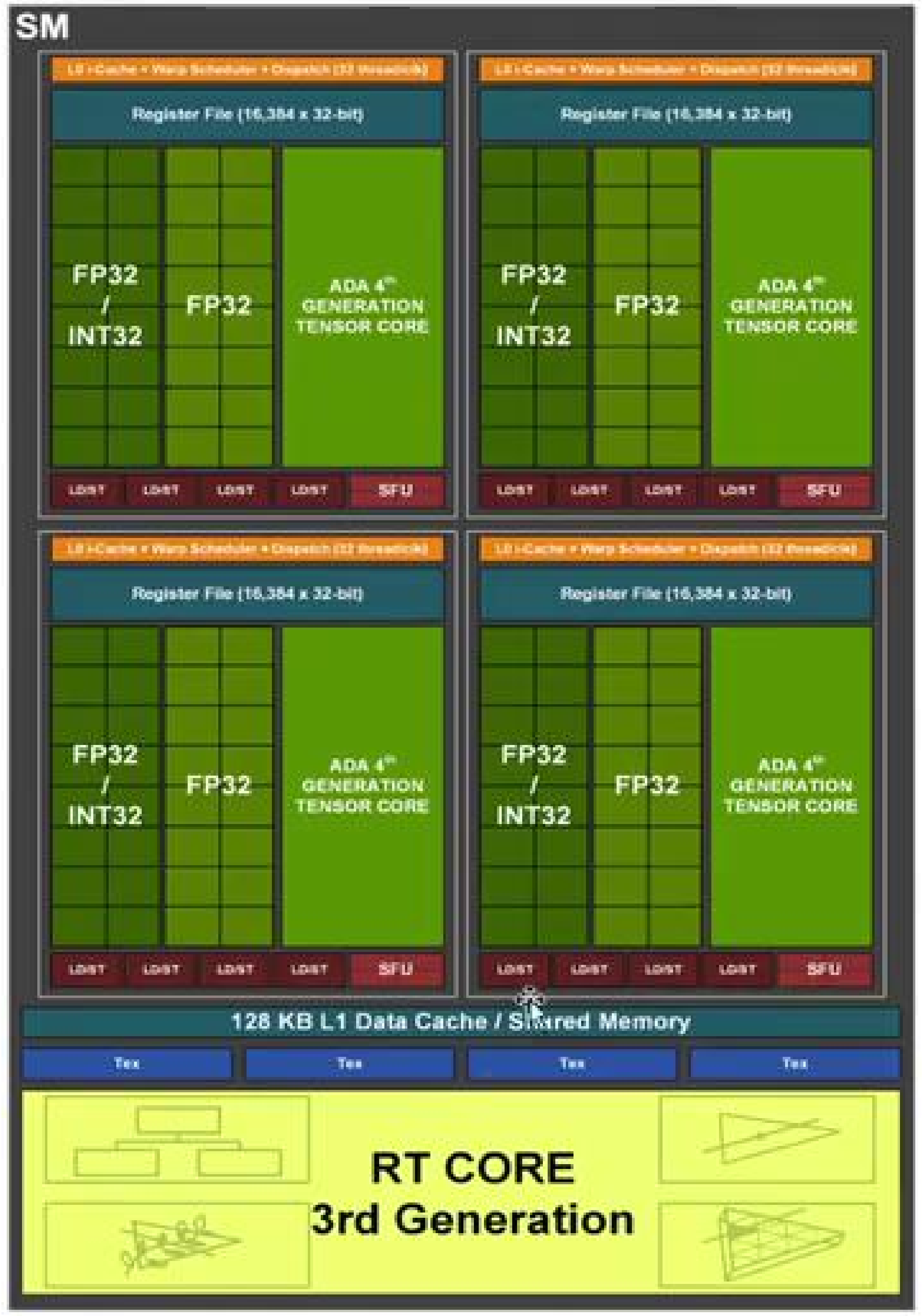


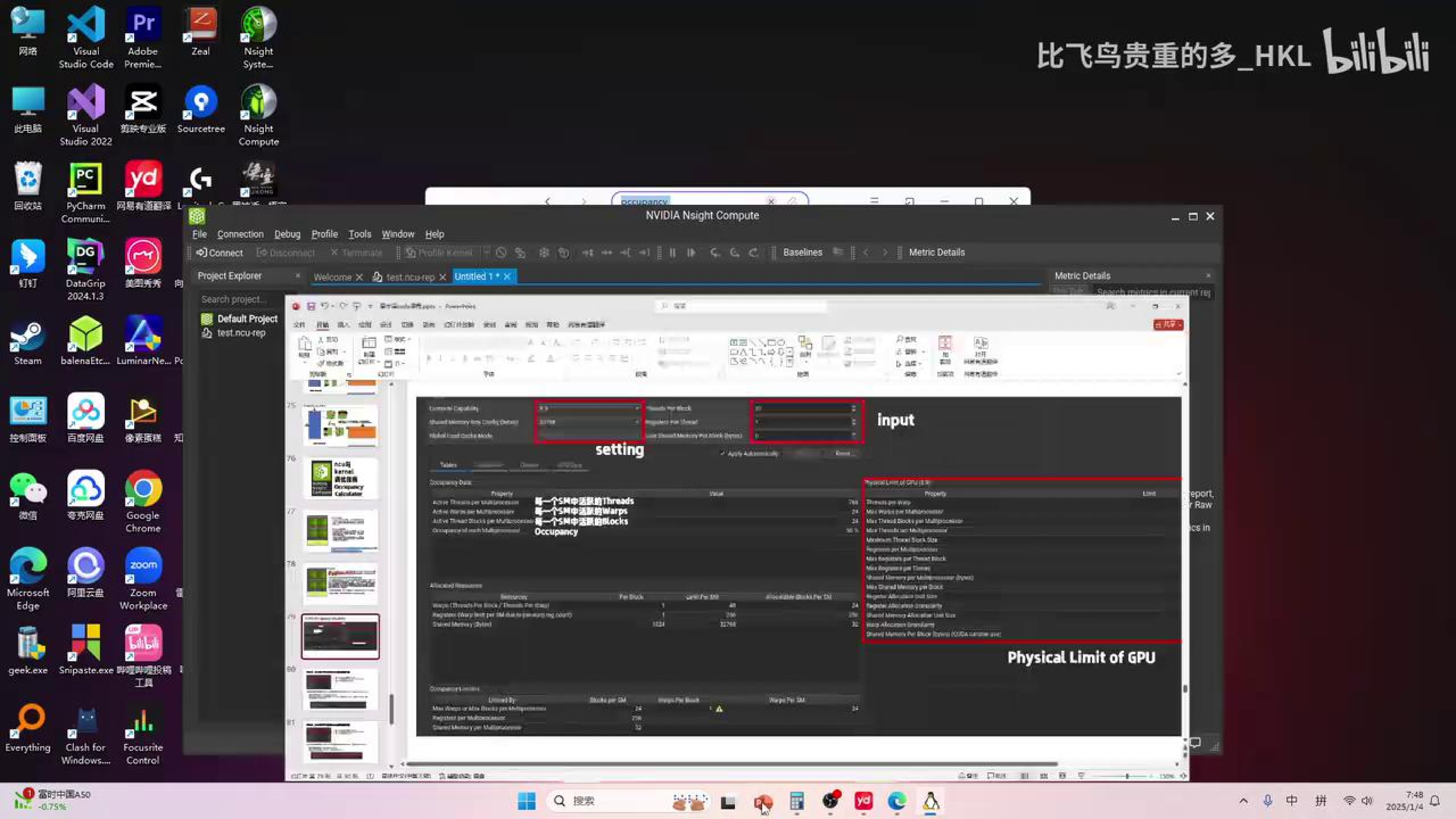
Figure 5. Ada Streaming Multiprocessor (SM)

occupancy初步理解

- occupancy中文翻译是占用率，一个SM中理论假如最大支持M个active warp（这里不理解什么是active warp可以参考这里）， 而实际上因为各种原因只能支持N个，这里N/M就是occupancy。

Active Warp（活跃线程束）：

- 在任何给定的时间点，GPU的SM（流处理器簇，Streaming Multiprocessors）可以同时处理多个warp。但是，由于资源限制（如执行单元、寄存器、共享内存等），并不是所有的warp都能同时活跃。
- Active warp指的是那些正在执行指令的warp。它们可能处于不同的执行阶段，比如正在执行计算任务、正在等待内存访问结果或正在执行控制流指令。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Metric Details
Project Explorer
Welcome
test.ncu-rep
Untitled 1 *
Default Project
test.ncu-rep
input
setting
每一个SM中活跃的Threads
每一个SM中活跃的Warps
每一个SM中活跃的Blocks
Occupancy
Physical Limit of GPU

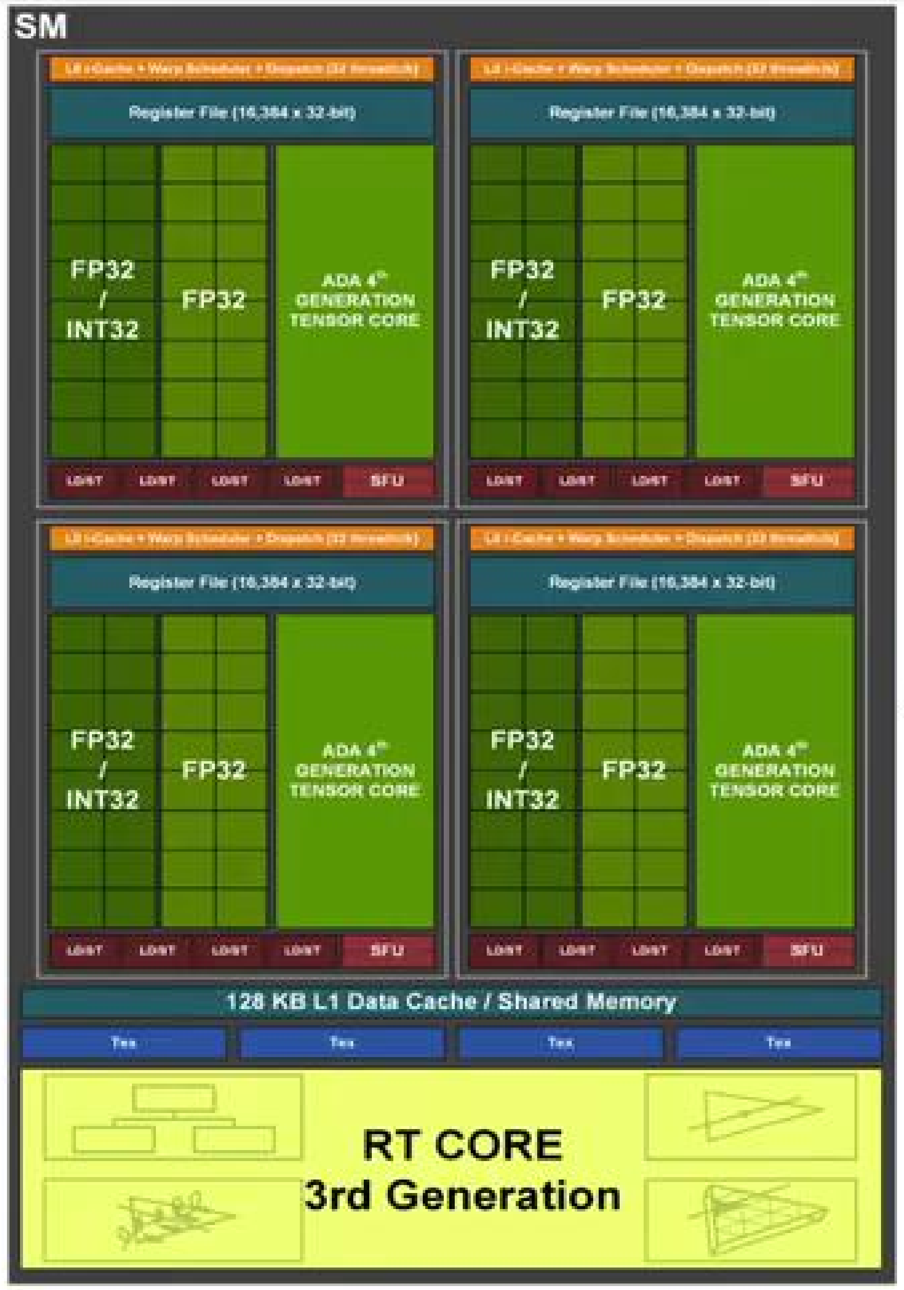


Figure 5. Ada Streaming Multiprocessor (SM)

以左侧图为例

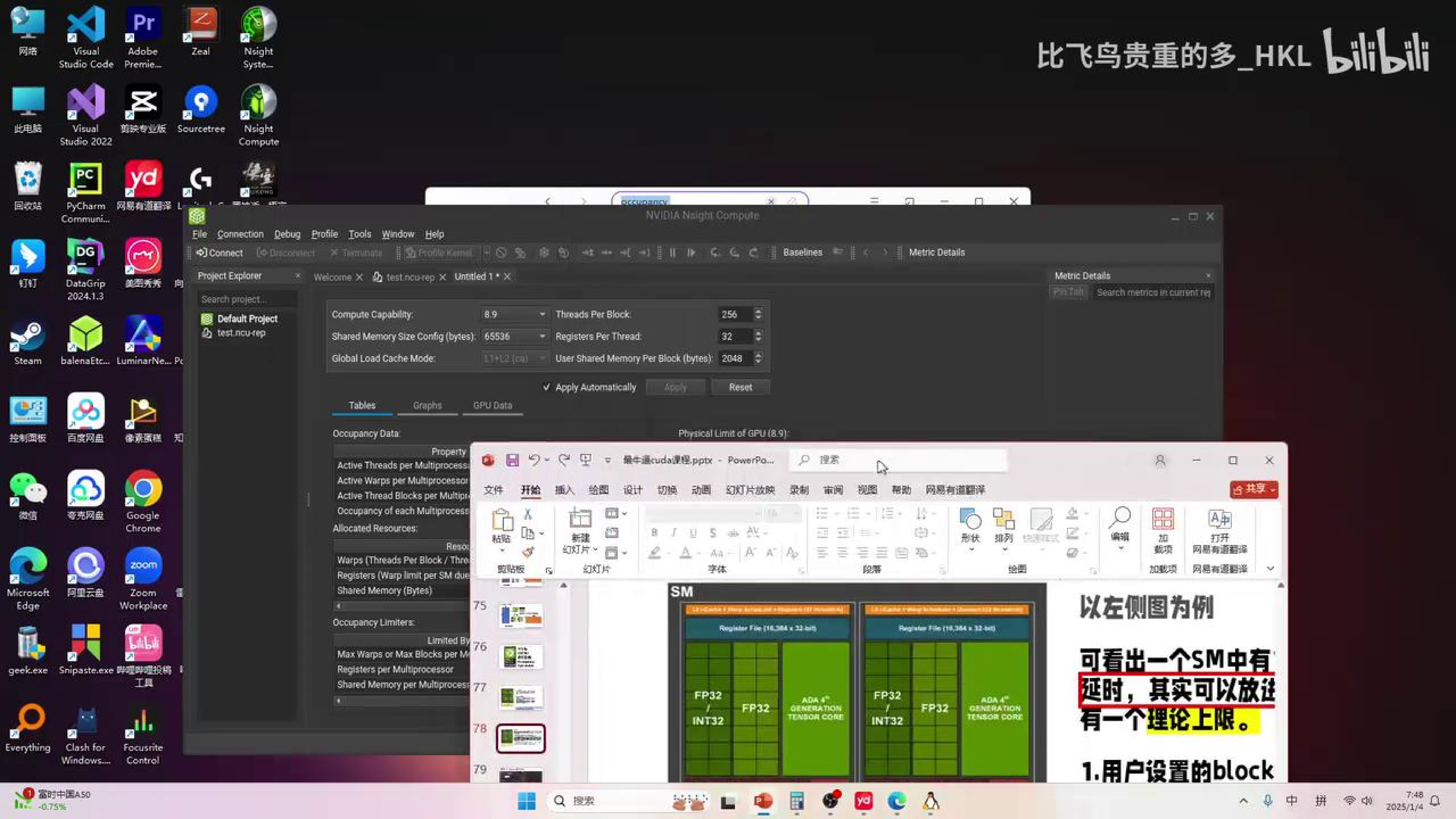
可看出一个SM中有128个Core，所以至少是可以放下128个Thread（4个w
延时，其实可以放进去更多warp（大于4个）根据显卡的架构不同，这个更
有一个理论上限。

1.用户设置的block的大小会影响实际SM上可以容纳的warp数量

2.CUDA隐藏延迟的手段就是当一个warp处于等待状态的时候会立刻切到下一
这个立刻就要求线程的切换要特别的迅速（0开销），这与CPU多线程有很大
进行thread切换的时候，是有开销的（要保留每个thread的上下文），但是
为它有大量的寄存器，每个thread的寄存器数据都是自己用所以就没有切换
自然也可以做到0开销，而与此同时每个SM中的寄存器数量是有上限的所以
能开启多少个warp的一个衡量指标

3.共享内存和寄存器类似，每个SM只有固定大小，所以也影响warp（block

123再结合木桶效应，取最小值就可以得到每个SM能容纳的warp数量。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File Connection Debug Profile Tools Window Help
Connect Disconnect Terminate Profile Kernel Baselines Metric Details
Project Explorer
Default Project
test.ncu-rep
Welcome test.ncu-rep Untitled 1 *
Compute Capability: 8.9
Threads Per Block: 256
Shared Memory Size Config (bytes): 65536
Registers Per Thread: 32
Global Load Cache Mode: L1+L2 (ca)
User Shared Memory Per Block (bytes): 2048
Apply Automatically Apply Reset
Tables Graphs GPU Data
Occupancy Data:
Physical Limit of GPU (8.9):
Property
Active Threads per Multiprocess
Active Warps per Multiprocessor
Active Thread Blocks per Multipr
Occupancy of each Multiprocess
Allocated Resources:
Warps (Threads Per Block / Thre
Registers (Warp limit per SM due
Shared Memory (Bytes)
Occupancy Limiters:
Limited By
Max Warps or Max Blocks per M
Registers per Multiprocessor
Shared Memory per Multiproces
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SM
Register File (16,384 x 32-bit)
FP32 / INT32
FP32
ADA 4th GENERATION TENSOR CORE
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可看出一个SM中有
延时，其实可以放进
有一个理论上限。
1.用户设置的block
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Project Explorer

Search project...

Default Project
test.ncu-rep

Welcome | test.ncu-rep | Untitled 1 *

Page: Details Result: 0 - 529 - transpose_naive | Add Baseline | Apply Rules | Occupancy Calculator | Copy as Image

	Result	Time	Cycles	Regs	GPU	SM Frequency	CC	Process
Current	529 - transpose_naive (64, 64, 1)x(8, 32, 1)	9.57 usecond	21,596	16	0 - NVIDIA GeForce RTX 4090	2.25 cycle/nsecond	8.9	[544] my_transpose

Statistics of the executed low-level assembly instructions (SASS). The instruction mix provides insight into the types and frequency of the executed instructions. A narrow mix of instruction types implies a dependency on few instruction pipelines, while others remain unused. Using multiple pipelines allows hiding latencies and enables parallel execution. Note that 'Instructions/Opcode' and 'Executed Instructions' are measured differently and can diverge if cycles are spent in system calls.

Executed Instructions [inst]	524288	Avg. Executed Instructions Per Scheduler [inst]	1024
Issued Instructions [inst]	551457	Avg. Issued Instructions Per Scheduler [inst]	1077.06

NVLink Topology

NVLink Topology diagram shows logical NVLink connections with transmit/receive throughput.

NVLink Tables

Detailed tables with properties for each NVLink.

NUMA Affinity

Non-uniform memory access (NUMA) affinities based on compute and memory distances for all GPUs.

Launch Statistics

Summary of the configuration used to launch the kernel. The launch configuration defines the size of the kernel grid, the division of the grid into blocks, and the GPU resources needed to execute the kernel. Choosing an efficient launch configuration maximizes device utilization.

Grid Size	4096	Function Cache Configuration	CachePreferNone
Registers Per Thread [register/thread]	16	Static Shared Memory Per Block [byte/block]	0
Block Size	256	Dynamic Shared Memory Per Block [byte/block]	0
Threads [thread]	1048576	Driver Shared Memory Per Block [Kbyte/block]	1.02
Waves Per SM	5.33	Shared Memory Configuration Size [Kbyte]	16.38

Occupancy

Occupancy is the ratio of the number of active warps per multiprocessor to the maximum number of possible active warps. Another way to view occupancy is the percentage of the hardware's ability to process warps that is actively in use. Higher occupancy does not always result in higher performance, however, low occupancy always reduces the ability to hide latencies, resulting in overall performance degradation. Large discrepancies between the theoretical and the achieved occupancy during execution typically indicates highly imbalanced workloads.

Theoretical Occupancy [%]	100	Block Limit Registers [block]	16
Theoretical Active Warps per SM [warp]	48	Block Limit Shared Mem [block]	16
Achieved Occupancy [%]	72.60	Block Limit Warps [block]	6
Achieved Active Warps Per SM [warp]	34.85	Block Limit SM [block]	24

Occupancy Limiters
Est. Speedup: 27.40%

This kernel's theoretical occupancy is not impacted by any block limit. The difference between calculated theoretical (100.0%) and measured achieved occupancy (72.6%) can be the result of warp scheduling overheads or workload imbalances during the kernel execution. Load imbalances can occur between warps within a block as well as across blocks of the same kernel. See the CUDA Best Practices Guide for more details on optimizing occupancy.

Source Counters

Source metrics, including branch efficiency and sampled warp stall reasons. Warp Stall Sampling metrics are periodically sampled over the kernel runtime. They indicate when warps were stalled and couldn't be scheduled. See the documentation for a description of all stall reasons. Only focus on stalls if the schedulers fail to issue every cycle.

Branch Instructions [inst]	32768	Branch Efficiency [%]	0
Branch Instructions Ratio [%]	0.06	Avg. Divergent Branches	0

Metric Details

Pin Tab | Search metrics in current report

launch__occupancy_limit_registers

Name	launch__occupancy_limit_registers
Unit	block
Value	16
Report	test.ncu-rep

Additional Information

Description:

Occupancy limit due to register usage.

Knowledgebase Entry:

registers: Each sub partition has a set of 32-bit registers, which are allocated by the HW in fixed-size chunks.

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NVIDIA Nsight Compute

File Connection Debug Profile Tools Window Help

Connect Disconnect Terminate Profile Kernel Baselines Metric Details

Project Explorer

Default Project
test.ncu-rep

Welcome | test.ncu-rep | Untitled 1 *

Page: Details Result: 0 - 529 - transpose_naive Add Baseline Apply Rules Occupancy Calculator Copy as Image

	Result	Time	Cycles	Regs	GPU	SM Frequency	CC	Process
Current	529 - transpose_naive (64, 64, 1)x(8, 32, 1)	9.57 usecond	21,596	16	0 - NVIDIA GeForce RTX 4090	2.25 cycle/nsecond	8.9	[544] my_transpose

average of 123.8 cycles between issuing two instructions.

Warp Stall Check the Warp Stall Sampling (All Samples) table for the top stall locations in your source based on sampling data. The Kernel Profiling Guide provides more details on each stall reason.

Instruction Statistics

Statistics of the executed low-level assembly instructions (SASS). The instruction mix provides insight into the types and frequency of the executed instructions. A narrow mix of instruction types implies a dependency on few instruction pipelines, while others remain unused. Using multiple pipelines allows hiding latencies and enables parallel execution. Note that 'Instructions/Opcode' and 'Executed Instructions' are measured differently and can diverge if cycles are spent in system calls.

Executed Instructions [inst]	524288	Avg. Executed Instructions Per Scheduler [inst]	1024
Issued Instructions [inst]	551457	Avg. Issued Instructions Per Scheduler [inst]	1077.06

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Summary of the configuration used to launch the kernel. The launch configuration defines the size of the kernel grid, the division of the grid into blocks, and the GPU resources needed to execute the kernel. Choosing an efficient launch configuration maximizes device utilization.

Grid Size	4096	Function Cache Configuration	CachePreferNone
Registers Per Thread [register/thread]	16	Static Shared Memory Per Block [byte/block]	0
Block Size	256	Dynamic Shared Memory Per Block [byte/block]	0
Threads [thread]	1048576	Driver Shared Memory Per Block [Kbyte/block]	1.02
Waves Per SM	5.33	Shared Memory Configuration Size [Kbyte]	16.38

Occupancy

Occupancy is the ratio of the number of active warps per multiprocessor to the maximum number of possible active warps. Another way to view occupancy is the percentage of the hardware's ability to process warps that is actively in use. Higher occupancy does not always result in higher performance, however, low occupancy always reduces the ability to hide latencies, resulting in overall performance degradation. Large discrepancies between the theoretical and the achieved occupancy during execution typically indicates highly imbalanced workloads.

Theoretical Occupancy [%]	100	Block Limit Registers [block]	16
Theoretical Active Warps per SM [warp]	48	Block Limit Shared Mem [block]	16
Achieved Occupancy [%]	72.60	Block Limit Warps [block]	6
Achieved Active Warps Per SM [warp]	34.85	Block Limit SM [block]	24

Occupancy Limiters
Est. Speedup: 27.40%
This kernel's theoretical occupancy is not impacted by any block limit. The difference between calculated theoretical (100.0%) and measured achieved occupancy (72.6%) can be the result of warp scheduling overheads or workload imbalances during the kernel execution. Load imbalances can occur between warps within a block as well as across blocks of the same kernel. See the CUDA Best Practices Guide for more details on optimizing occupancy.

Metric Details

Pin Tab | Search metrics in current report

launch__occupancy_limit_shared_mem

Name	launch__occupancy_limit_shared_...
Unit	block
Value	16
Report	test.ncu-rep

Additional Information

Description:

Occupancy limit due to shared memory usage.

Knowledgebase Entry:

shared: Shared memory is located on chip, so it has much higher bandwidth and much lower latency than either local or global memory. Shared memory can be shared across a compute CTA.

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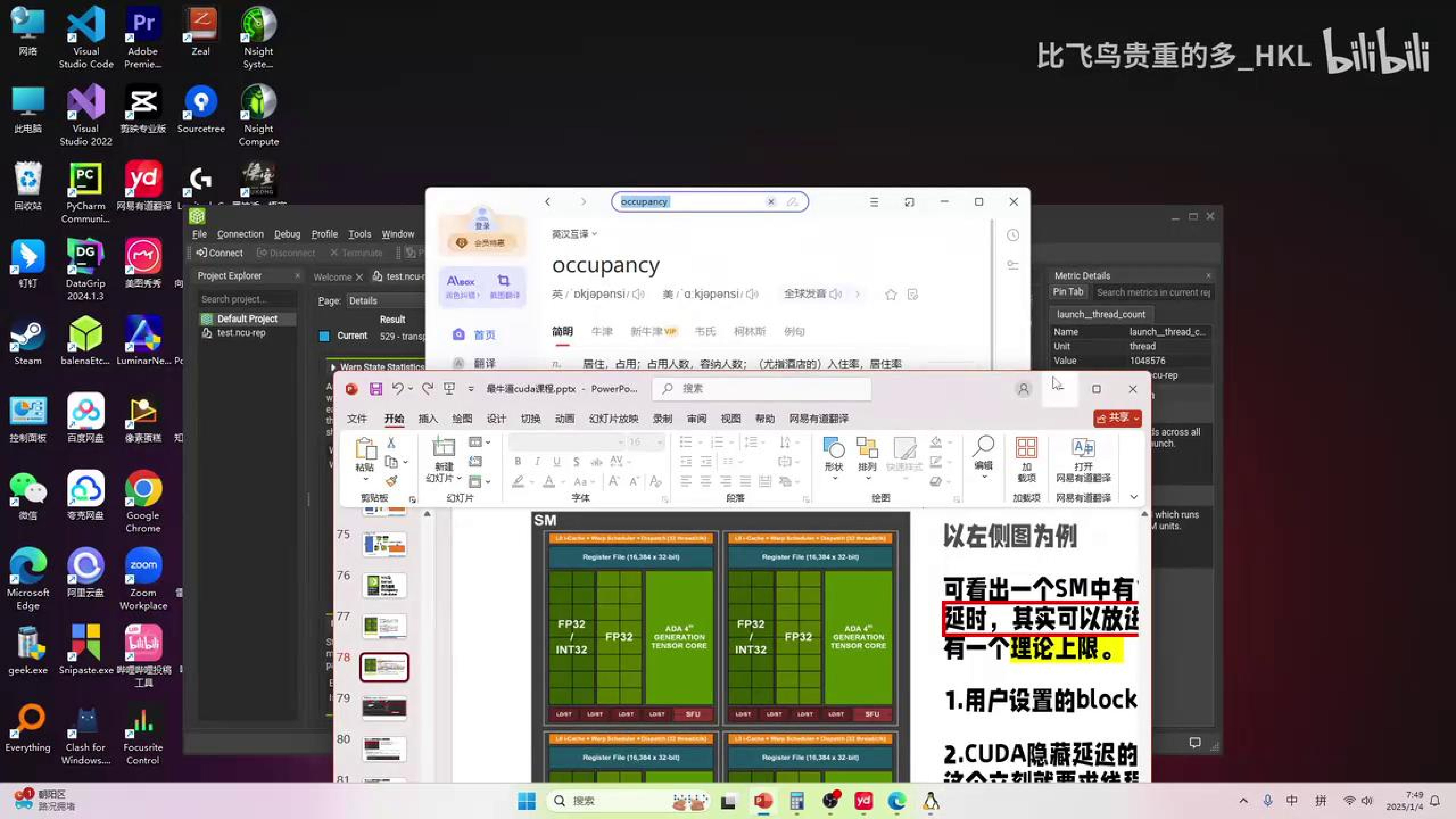

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Warp State Statistics
Metric Details
Pin Tab
launch__thread_count
Name launch__thread_c...
Unit thread
Value 1048576
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SM
Register File (16,384 x 32-bit)
FP32 / INT32
FP32
ADA 4th GENERATION TENSOR CORE
SFU
以左侧图为例
可看出一个SM中有
延时，其实可以放
有一个理论上限。
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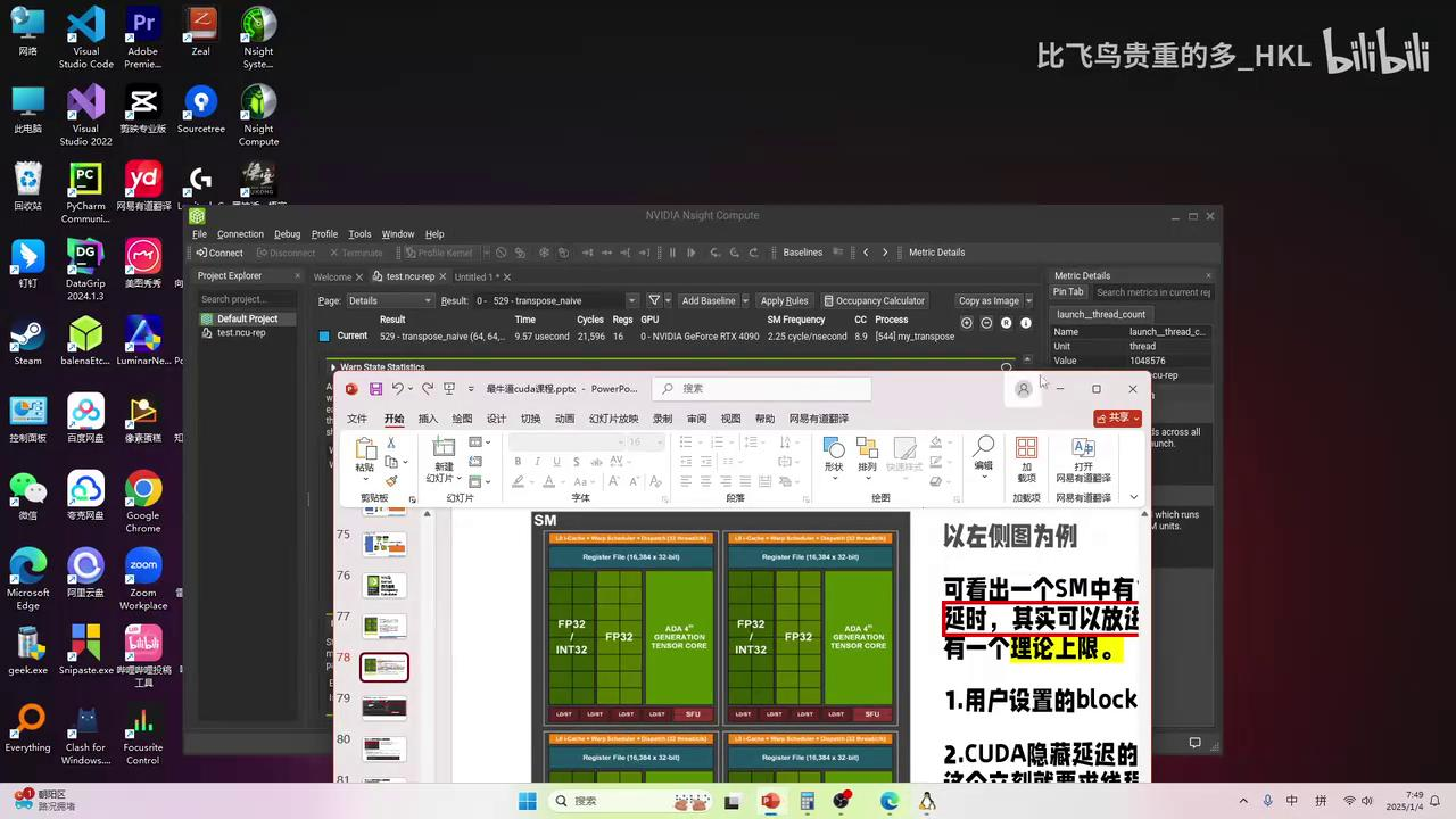
2.CUDA隐藏延迟的



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Connect Disconnect Terminate Profile Kernel Baselines Metric Details
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Default Project
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Page: Details Result: 0 - 529 - transpose_naive
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Warp State Statistics
Metric Details
Pin Tab
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Name launch__thread_c...
Unit thread
Value 1048576
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剪贴板 幻灯片 字体 段落 绘图 加载项 网易有道翻译
SM
Register File (16,384 x 32-bit)
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1.用户设置的block
2.CUDA隐藏延迟的
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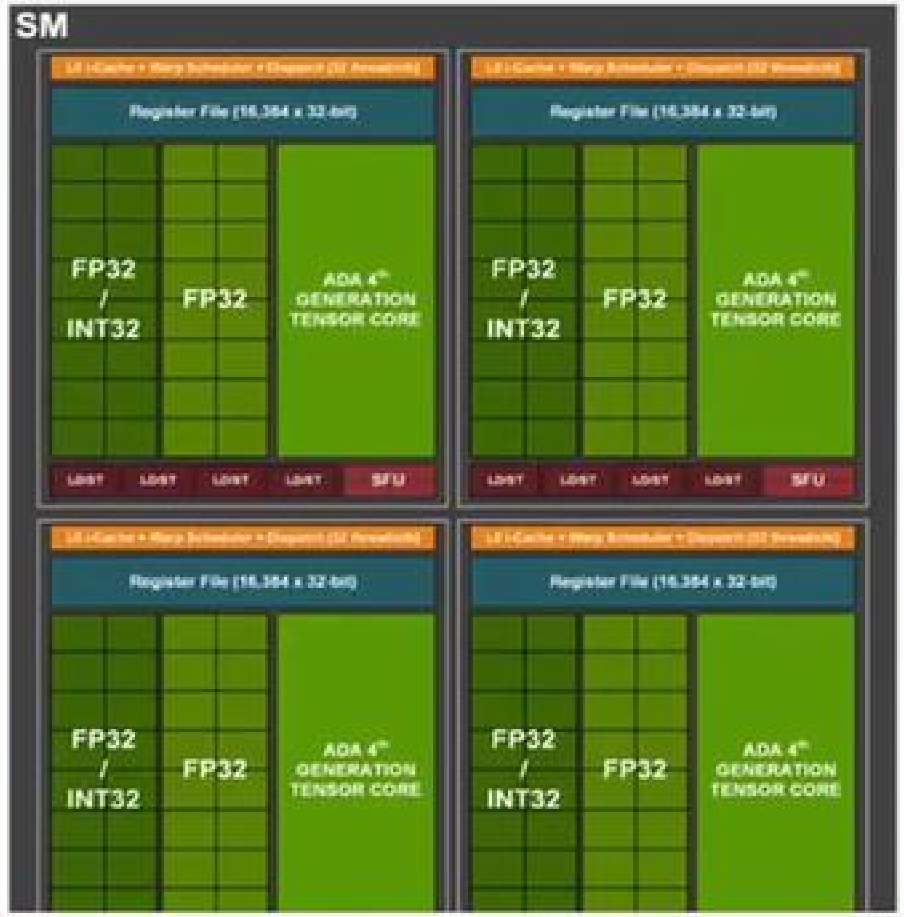


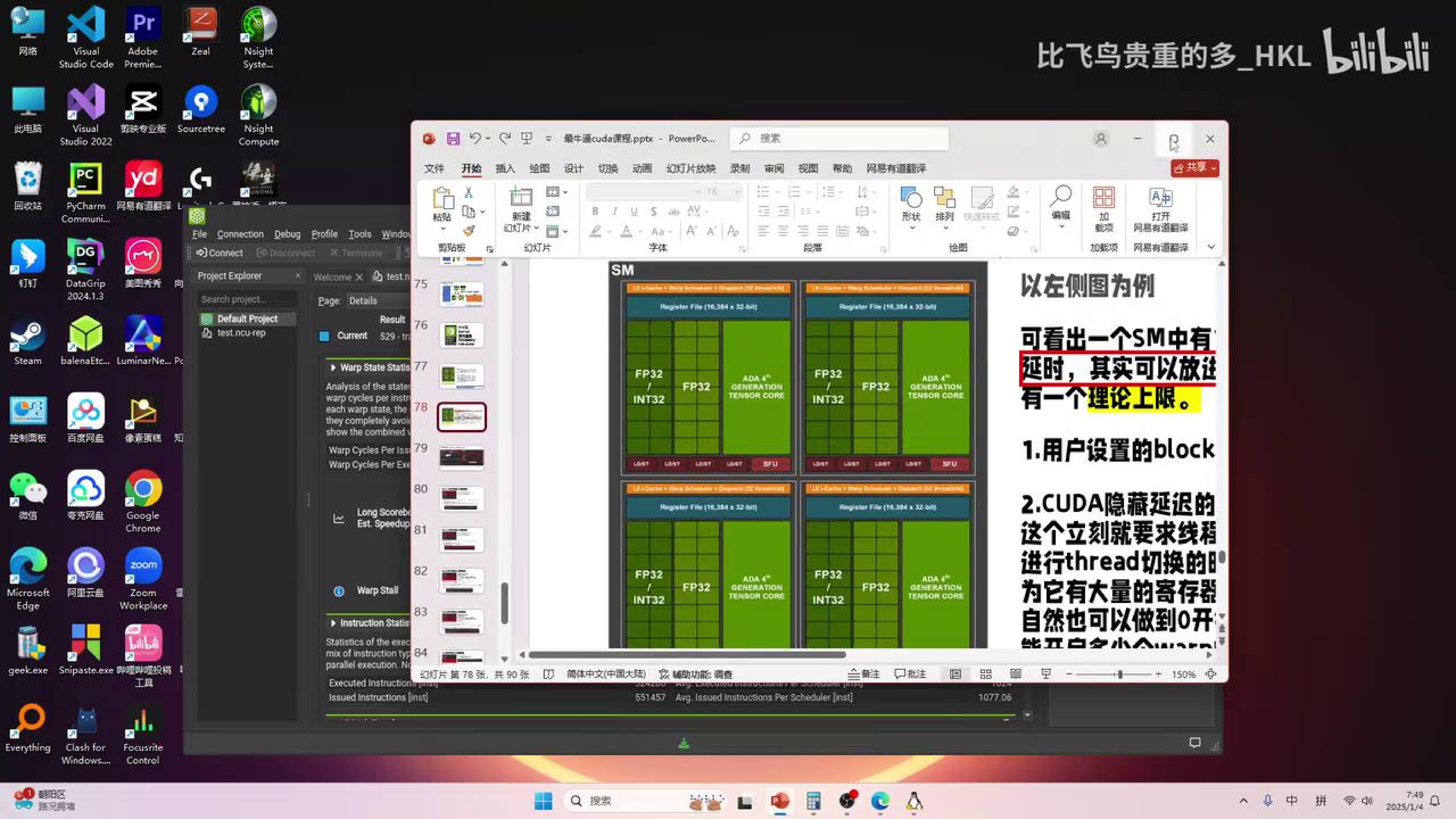
以左侧图为例

可看出一个SM中有
延时，其实可以放进
有一个理论上限。

1.用户设置的block

2.CUDA隐藏延迟的
这个立刻就要求线程
进行thread切换的
为它有大量的寄存器
自然也可以做到0开
能开启多少个warp

Executed Instructions [inst]	524288	Avg. Executed Instructions Per Scheduler [inst]	1024
Issued Instructions [inst]	551457	Avg. Issued Instructions Per Scheduler [inst]	1077.06



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SM
Register File (16,384 x 32-bit)
FP32 / INT32
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ADA 4th GENERATION TENSOR CORE
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可看出一个SM中有
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这个立刻就要求线程
进行thread切换的
为它有大量的寄存器
自然也可以做到0开
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Project Explorer
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test.ncu-rep
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Long Scorebo
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Warp Stall
Instruction Statis
Issued Instructions [inst]
551457
Avg. Issued Instructions Per Scheduler [inst]
1077.06

Figure 5. Ada Streaming Multiprocessor (SM)

以左侧图为例

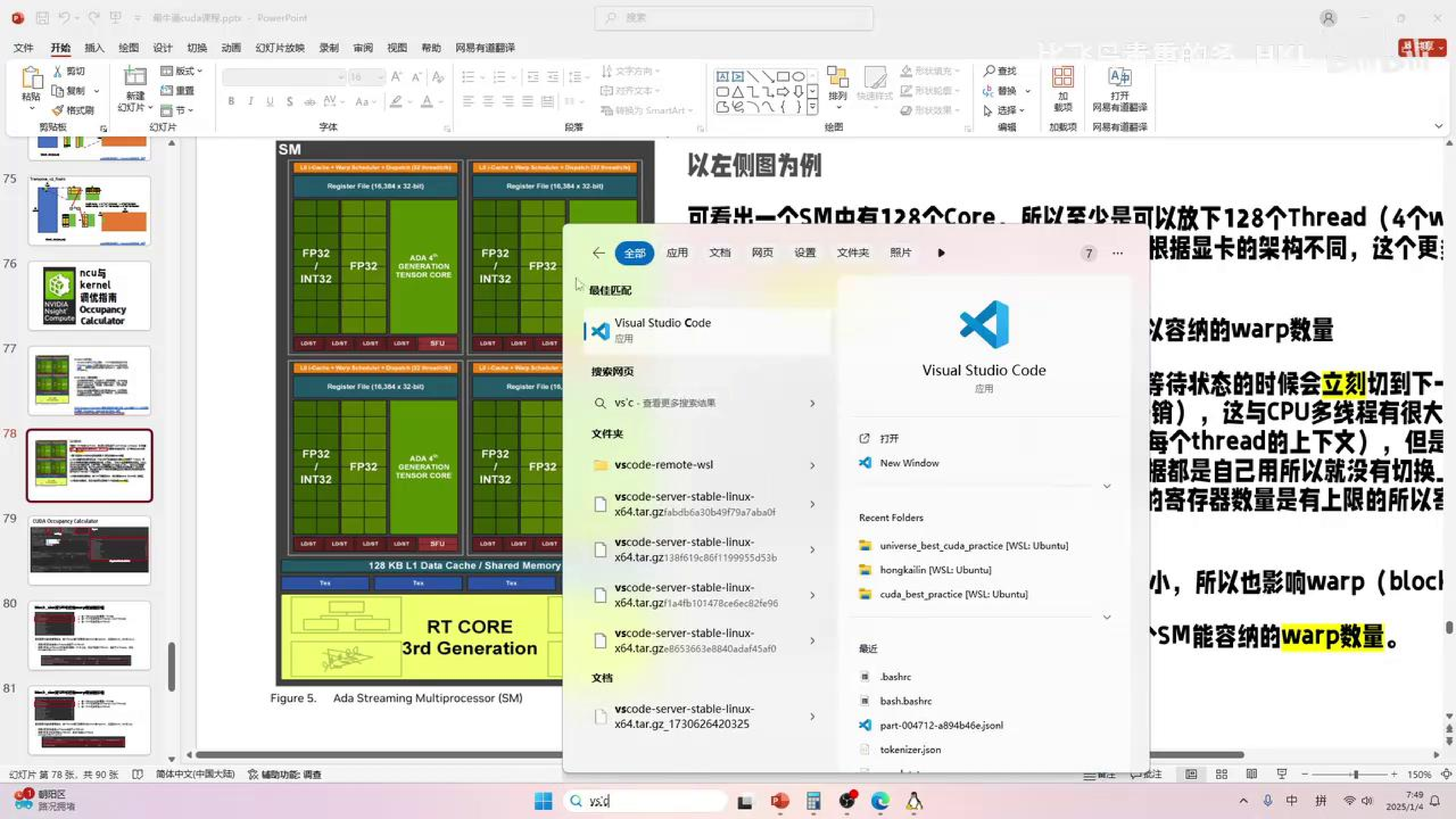
可看出一个SM中有128个Core，所以至少是可以放下128个Thread（4个w
根据显卡的架构不同，这个更

以容纳的warp数量

等待状态的时候会**立刻**切到下一
销），这与CPU多线程有很大
每个thread的上下文），但是
据都是自己用所以就没有切换
的寄存器数量是有上限的所以

小，所以也影响warp（block

个SM能容纳的**warp数量**。



第十章cuda课程.pptx - PowerPoint
文件 开始 插入 绘图 设计 切换 动画 幻灯片放映 录制 审阅 视图 帮助 网易有道翻译
SM
Register File (16,384 x 32-bit)
FP32 / INT32
FP32
ADA 4th GENERATION TENSOR CORE
SFU
128 KB L1 Data Cache / Shared Memory
Tex
RT CORE 3rd Generation
Figure 5. Ada Streaming Multiprocessor (SM)
以左侧图为例
可看出一个SM中有128个Core，所以至少是可以放下128个Thread（4个w
根据显卡的架构不同，这个更
以容纳的warp数量
等待状态的时候会立刻切到下一
销），这与CPU多线程有很大
每个thread的上下文），但是
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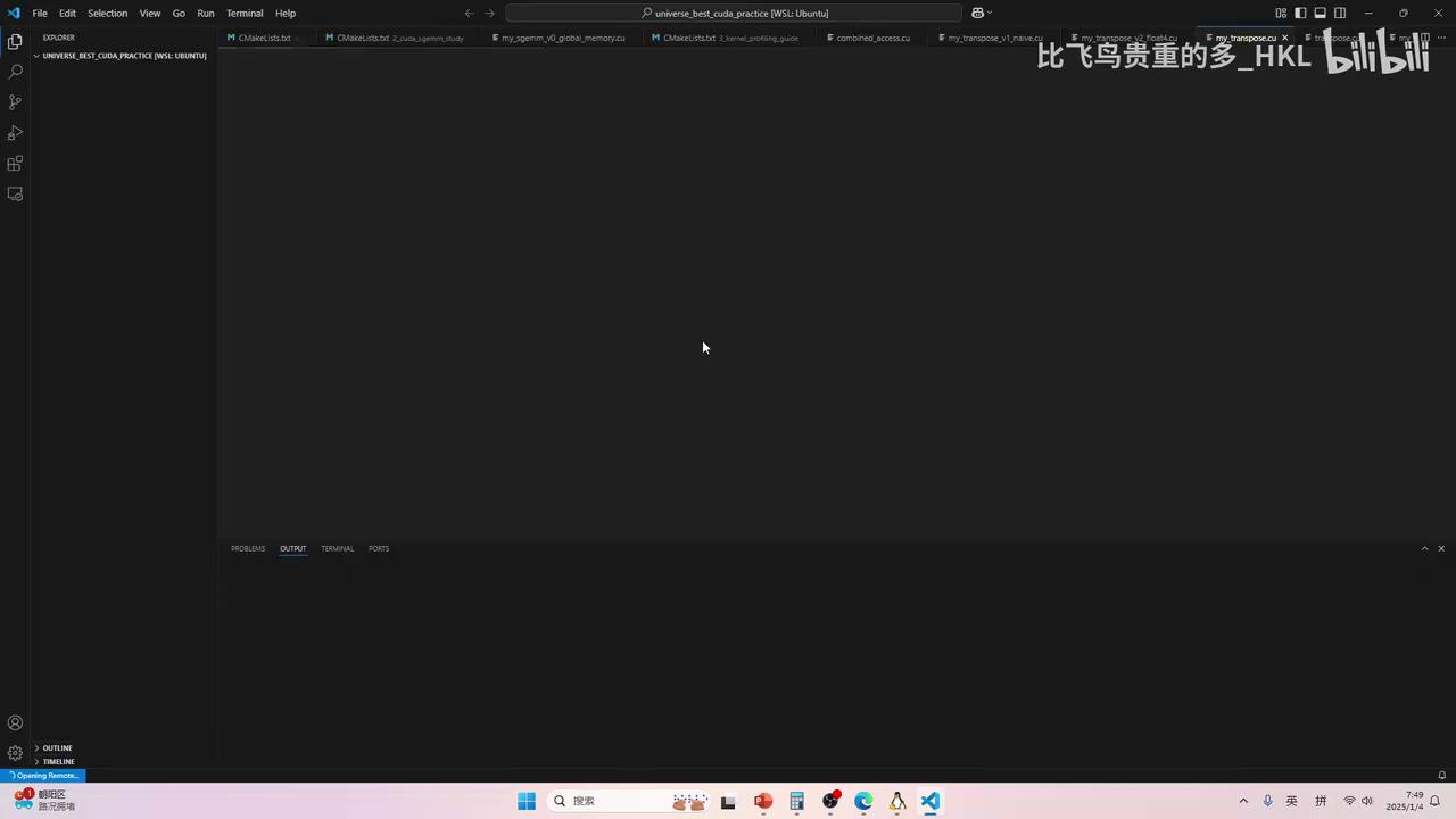
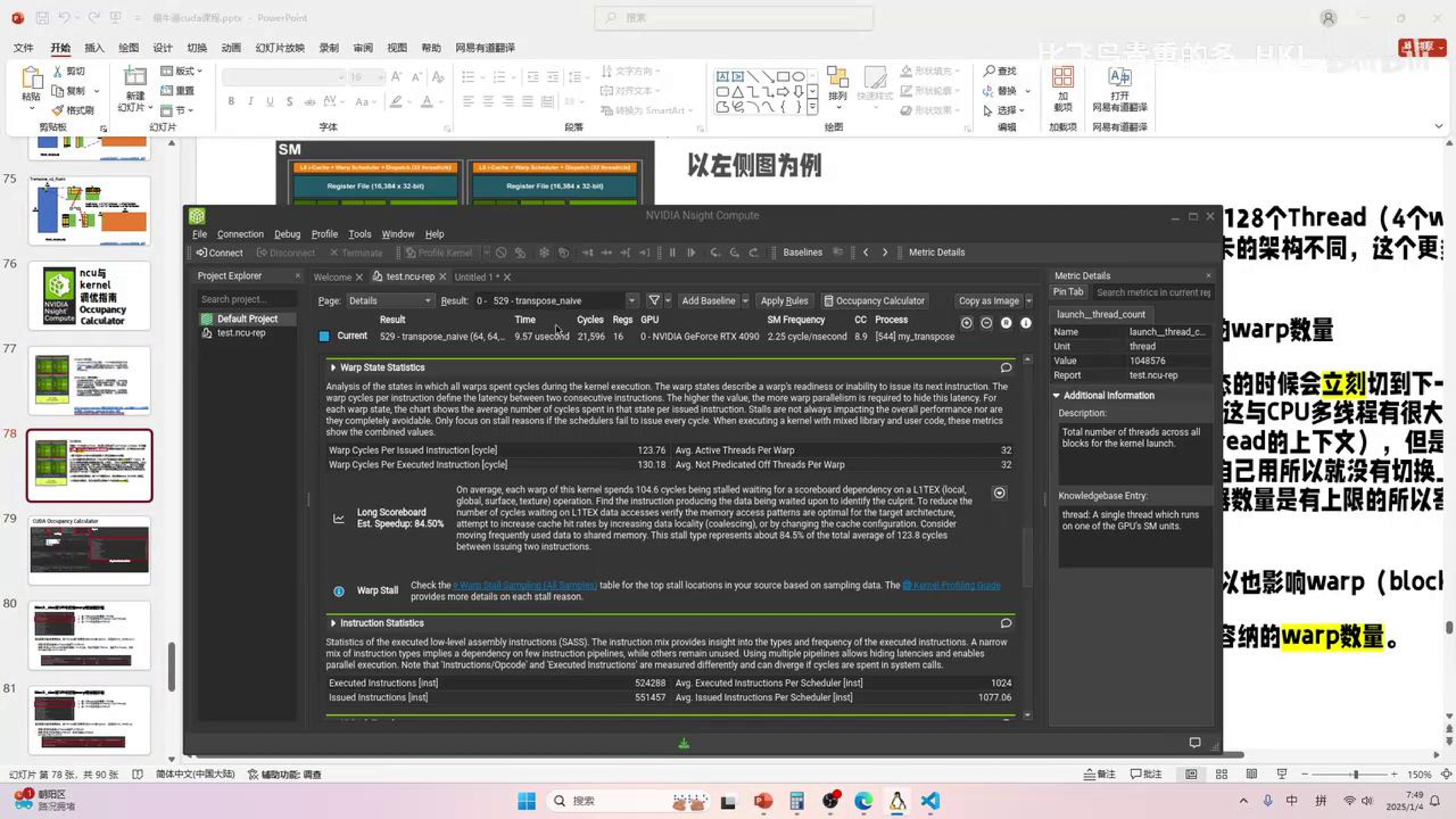
的寄存器数量是有上限的所以
小，所以也影响warp（block
个SM能容纳的warp数量。
全部 应用 文档 网页 设置 文件夹 照片
最佳匹配
Visual Studio Code
应用
搜索网页
vs'c - 查看更多搜索结果
文件夹
vscode-remote-wsl
vscode-server-stable-linux-x64.tar.gzfabdb6a30b49f79a7aba0f
vscode-server-stable-linux-x64.tar.gz138f619c86f1199955d53b
vscode-server-stable-linux-x64.tar.gzf1a4fb101478ce6ec82fe96
vscode-server-stable-linux-x64.tar.gze8653663e8840adaf45af0
文档
vscode-server-stable-linux-x64.tar.gz_1730626420325
打开
New Window
Recent Folders
universe_best_cuda_practice [WSL: Ubuntu]
hongkailin [WSL: Ubuntu]
cuda_best_practice [WSL: Ubuntu]
最近
.bashrc
bash.bashrc
part-004712-a894b46e.jsonl
tokenizer.json
幻灯片 第 78 张，共 90 张
简体中文(中国大陆)
vs'c
7:49
2025/1/4


File Edit Selection View Go Run Terminal Help
universe_best_cuda_practice [WSL: Ubuntu]
EXPLORER
UNIVERSE_BEST_CUDA_PRACTICE [WSL: UBUNTU]
CMakeLists.txt
CMakeLists.txt 2_cuda_sgemm_study
my_sgemm_v0_global_memory.cu
CMakeLists.txt 3_kernel_profiling_guide
combined_access.cu
my_transpose_v1_naive.cu
my_transpose.cu
比飞鸟贵重的多_HKL
PROBLEMS
OUTPUT
TERMINAL
PORTS
OUTLINE
TIMELINE
Opening Remote...
朝阳区
搜索
英
拼
7:49
2025/1/4



SM
L0 i-Cache + Warp Scheduler + Dispatch (32 thread/clk)
Register File (16,384 x 32-bit)
以左侧图为例
128个Thread（4个w
卡的架构不同，这个更
的warp数量
态的时候会立刻切到下-
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read的上下文），但是
自己用所以就没有切换
器数量是有上限的所以
以也影响warp（block
容纳的warp数量。
NVIDIA Nsight Compute
File Connection Debug Profile Tools Window Help
Connect Disconnect Terminate Profile Kernel Baselines Metric Details
Project Explorer
Default Project
test.ncu-rep
Welcome test.ncu-rep Untitled 1 *
Page: Details Result: 0 - 529 - transpose_naive Add Baseline Apply Rules Occupancy Calculator Copy as Image
Result Time Cycles Regs GPU SM Frequency CC Process
Current 529 - transpose_naive (64, 64,... 9.57 usecond 21,596 16 0 - NVIDIA GeForce RTX 4090 2.25 cycle/nsecond 8.9 [544] my_transpose
Warp State Statistics
Analysis of the states in which all warps spent cycles during the kernel execution. The warp states describe a warp's readiness or inability to issue its next instruction. The warp cycles per instruction define the latency between two consecutive instructions. The higher the value, the more warp parallelism is required to hide this latency. For each warp state, the chart shows the average number of cycles spent in that state per issued instruction. Stalls are not always impacting the overall performance nor are they completely avoidable. Only focus on stall reasons if the schedulers fail to issue every cycle. When executing a kernel with mixed library and user code, these metrics show the combined values.
Warp Cycles Per Issued Instruction [cycle] 123.76 Avg. Active Threads Per Warp 32
Warp Cycles Per Executed Instruction [cycle] 130.18 Avg. Not Predicated Off Threads Per Warp 32
Long Scoreboard Est. Speedup: 84.50%
On average, each warp of this kernel spends 104.6 cycles being stalled waiting for a scoreboard dependency on a L1TEX (local, global, surface, texture) operation. Find the instruction producing the data being waited upon to identify the culprit. To reduce the number of cycles waiting on L1TEX data accesses verify the memory access patterns are optimal for the target architecture, attempt to increase cache hit rates by increasing data locality (coalescing), or by changing the cache configuration. Consider moving frequently used data to shared memory. This stall type represents about 84.5% of the total average of 123.8 cycles between issuing two instructions.
Warp Stall Check the Warp Stall Sampling (All Samples) table for the top stall locations in your source based on sampling data. The Kernel Profiling Guide provides more details on each stall reason.
Instruction Statistics
Statistics of the executed low-level assembly instructions (SASS). The instruction mix provides insight into the types and frequency of the executed instructions. A narrow mix of instruction types implies a dependency on few instruction pipelines, while others remain unused. Using multiple pipelines allows hiding latencies and enables parallel execution. Note that 'Instructions/Opcode' and 'Executed Instructions' are measured differently and can diverge if cycles are spent in system calls.
Executed Instructions [inst] 524288 Avg. Executed Instructions Per Scheduler [inst] 1024
Issued Instructions [inst] 551457 Avg. Issued Instructions Per Scheduler [inst] 1077.06
Metric Details
Pin Tab
launch__thread_count
Name launch__thread_c...
Unit thread
Value 1048576
Report test.ncu-rep
Additional Information
Description:
Total number of threads across all blocks for the kernel launch.
Knowledgebase Entry:
thread: A single thread which runs on one of the GPU's SM units.
幻灯片 第 78 张，共 90 张

NVIDIA Nsight Compute

File Connection Debug Profile Tools Window Help

Connect Disconnect Terminate Profile Kernel Baselines Metric Details

Project Explorer

Search project...

Default Project

test.ncu-rep

Welcome test.ncu-rep Untitled 1 * Untitled 2 *

Compute Capability:	8.9	Threads Per Block:	256
Shared Memory Size Config (bytes):	16384	Registers Per Thread:	16
Global Load Cache Mode:	L1+L2 (ca)	User Shared Memory Per Block (bytes):	0

✓ Apply Automatically Apply Reset

Tables Graphs GPU Data

Occupancy Data:

Property	Value
Active Threads per Multiprocessor	1536
Active Warps per Multiprocessor	48
Active Thread Blocks per Multiprocessor	6
Occupancy of each Multiprocessor	100 %

Allocated Resources:

Resources	Per Block
Warps (Threads Per Block / Threads Per Warp)	
Registers (Warp limit per SM due to per-warp reg count)	
Shared Memory (Bytes)	

Occupancy Limiters:

Limited By	Blocks per SM
Max Warps or Max Blocks per Multiprocessor	6
Registers per Multiprocessor	16
Shared Memory per Multiprocessor	16

Physical Limit of GPU (8.9):

Property
Threads per Warp
Max Warps per Multiprocessor
Max Thread Blocks per Multiprocessor
Max Threads per Multiprocessor
Maximum Thread Block Size
Registers per Multiprocessor
Max Registers per Thread Block
Max Registers per Thread
Shared Memory per Multiprocessor (bytes)
Max Shared Memory per Block
Register Allocation Unit Size
Register Allocation Granularity
Shared Memory Allocation Unit Size
Warp Allocation Granularity
Shared Memory Per Block (bytes) (CUDA runtime use)

Metric Details

Pin Tab Search metrics in current report

launch__thread_count

Name	launch__thread_c...
Unit	thread
Value	1048576
Report	test.ncu-rep

Additional Information

Description:

Total number of threads across all blocks for the kernel launch

Knowledgebase Entry:

thread: A single thread which runs on one of the GPU's SM units.

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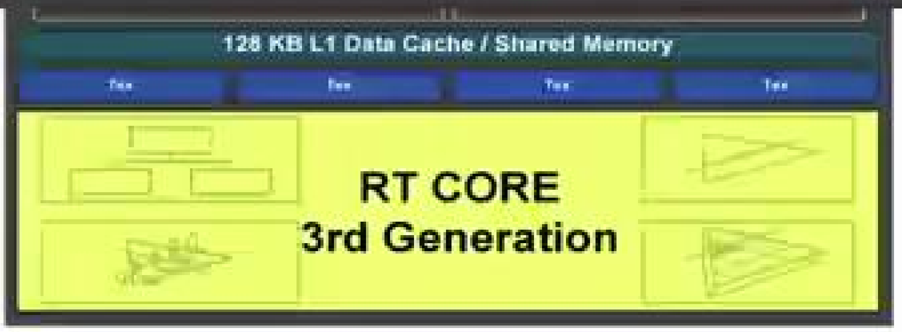

Figure 5. Ada Streaming Multiprocessor (SM)

3.共享内存和寄存器类似，每个SM只有固定大小，所以也影响warp（block

123再结合木桶效应，取最小值就可以得到每个SM能容纳的warp数量。

幻灯片 第 78 张，共 90 张 简体中文(中国大陆) 辅助功能: 调查 150%

NVIDIA Nsight Compute

File Connection Debug Profile Tools Window Help

Connect Disconnect Terminate Profile Kernel Baselines Metric Details

比飞鸟贵重的多_HKL bilibili

Project Explorer

Search project...

Default Project
test.ncu-rep

Welcome | test.ncu-rep | Untitled 1 * | Untitled 2 *

Compute Capability:	8.9	Threads Per Block:	256
Shared Memory Size Config (bytes):	16384	Registers Per Thread:	2
Global Load Cache Mode:	L1+L2 (ca)	User Shared Memory Per Block (bytes):	0

✓ Apply Automatically | Apply | Reset

Tables | Graphs | GPU Data

Occupancy Data:

Property	Value
Active Threads per Multiprocessor	1536
Active Warps per Multiprocessor	48
Active Thread Blocks per Multiprocessor	6
Occupancy of each Multiprocessor	100 %

Allocated Resources:

Resources	Per Block	Limit Per SM
Warps (Threads Per Block / Threads Per Warp)	8	48
Registers (Warp limit per SM due to per-warp reg count)	8	25
Shared Memory (Bytes)	1024	1638

Occupancy Limiters:

Limited By	Blocks per SM	Warps Per Block	W
Max Warps or Max Blocks per Multiprocessor	6		
Registers per Multiprocessor	32		
Shared Memory per Multiprocessor	16		

Physical Limit of GPU (8.9):

Property	Limit
Threads per Warp	32
Max Warps per Multiprocessor	48
Max Thread Blocks per Multiprocessor	24
Max Threads per Multiprocessor	1536
Maximum Thread Block Size	1024
Registers per Multiprocessor	65536
Max Registers per Thread Block	65536
Max Registers per Thread	255
Shared Memory per Multiprocessor (bytes)	16384
Max Shared Memory per Block	16384
Register Allocation Unit Size	256
Register Allocation Granularity	warp
Shared Memory Allocation Unit Size	128
Warp Allocation Granularity	4
Shared Memory Per Block (bytes) (CUDA runtime use)	1024

Metric Details

Pin Tab | Search metrics in current report

launch__thread_count

Name	launch__thread_count
Unit	thread
Value	1048576
Report	test.ncu-rep

Additional Information

Description:

Total number of threads across all blocks for the kernel launch.

Knowledgebase Entry:

thread: A single thread which runs on one of the GPU's SM units.

自选股 观点 | 搜索 | 中 拼 | 7:50 2025/1/4

block_size对SM可启动warp数量的影响

Physical Limit of GPU (8.9):

Property	Limit
Threads per Warp	32
Max Warps per Multiprocessor	48
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Max Threads per Multiprocessor	1536
Maximum Thread Block Size	1024
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Max Registers per Thread Block	65536
Max Registers per Thread	255
Shared Memory per Multiprocessor (bytes)	16384
Max Shared Memory per Block	16384
Register Allocation Unit Size	256
Register Allocation Granularity	warp
Shared Memory Allocation Unit Size	128
Warp Allocation Granularity	4
Shared Memory Per Block (bytes) (CUDA runtime use)	1024

1. 每一个Block必须分配到一个SM中
2. 每一个SM最多可放48个warp(1536个thread)
3. 每一个SM最多可放24个Block

假设需要计算的数据足够多，每个thread都只使用很少的smem和register，设置的Block_Size为(32,5)

- 根据2可得最多能放48个warp相当于9.6个Block
- 根据1可得0.6个Block是不能被分配到一个SM上的，所以只能放9个Block，相当于45个warp，所以occupany为0.9375=48/45

Allocated Resources:

Resources	Per Block	Limit Per SM	Allocatable Blocks Per SM
Warps (Threads Per Block / Threads Per Warp)	5	48	9
Registers (Warp limit per SM due to per-warp reg count)	5	256	51
Shared Memory (Bytes)	1024	65536	64

_size对SM可启动warp数量的影响

	Limit
	32
	48
cessor	24
	1536
	1024
	65536
	65536
	255
sor (bytes)	16384
	16384
	256
	warp
Size	128
	4
s) (CUDA runtime use)	1024

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ny为0.9375=45/48

幻灯片 第 80 张，共 90 张

block_size对SM可启动warp数量的影响

Physical Limit of GPU (8.9):

Property	Limit
Threads per Warp	32
Max Warps per Multiprocessor	48
Max Thread Blocks per Multiprocessor	24
Max Threads per Multiprocessor	1536
Maximum Thread Block Size	1024
Registers per Multiprocessor	65536
Max Registers per Thread Block	65536
Max Registers per Thread	255
Shared Memory per Multiprocessor (bytes)	16384
Max Shared Memory per Block	16384
Register Allocation Unit Size	256
Register Allocation Granularity	warp
Shared Memory Allocation Unit Size	128
Warp Allocation Granularity	4
Shared Memory Per Block (bytes) (CUDA runtime use)	1024

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3. 每一个SM最多可放24个Block

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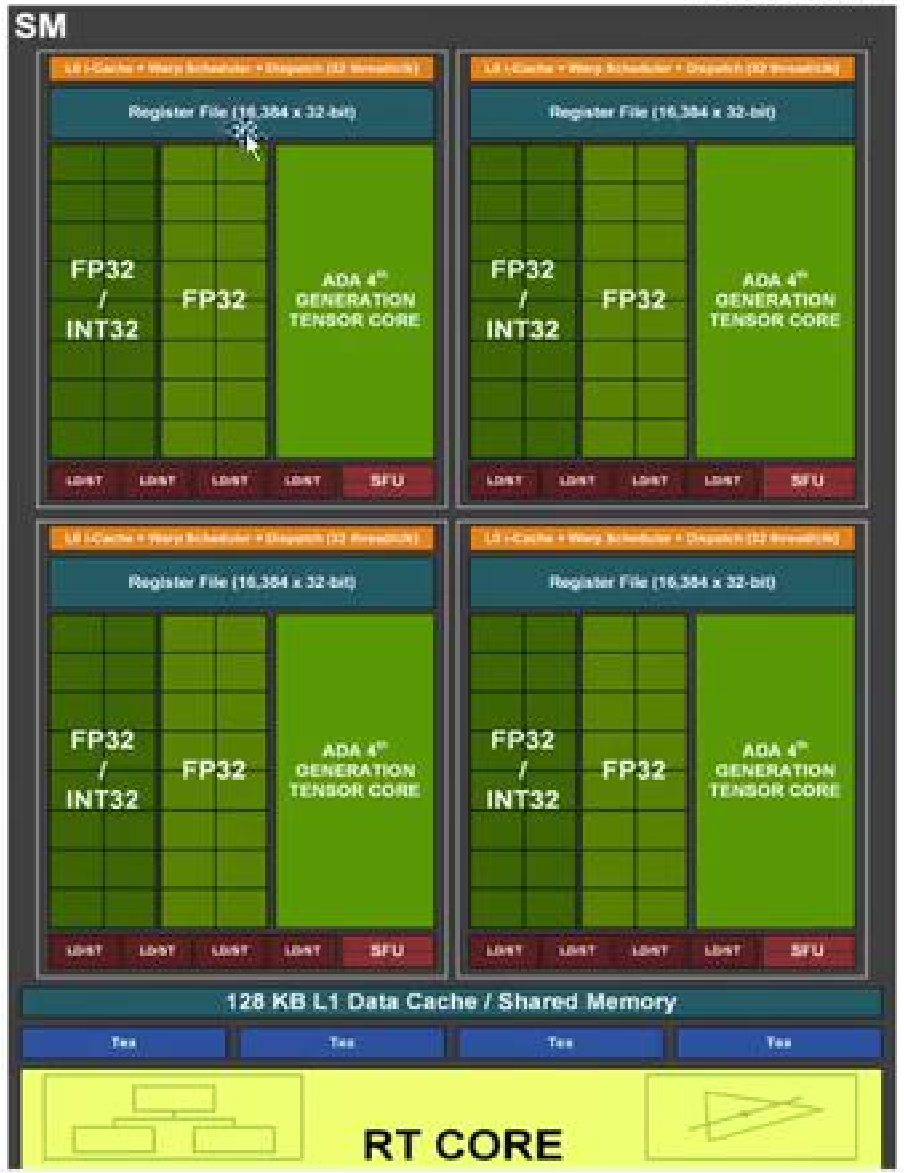
Ada GPU Architecture In-Depth



以左侧图为例

**可看出一个SM中有128个Core，所以至少是可以放下128个Thread（4个w
延时，其实可以放进去更多warp（大于4个）根据显卡的架构不同，这个更
有一个理论上限。**

1.用户设置的block的大小会影响实际SM上可以容纳的warp数量

**2.CUDA隐藏延迟的手段就是当一个warp处于等待状态的时候会立刻切到下一
这个立刻就要求线程的切换要特别的迅速（0开销），这与CPU多线程有很大
进行thread切换的时候，是有开销的（要保留每个thread的上下文），但是
为它有大量的寄存器，每个thread的寄存器数据都是自己用所以就没有切换
自然也可以做到0开销，而与此同时每个SM中的寄存器数量是有上限的所以
能开启多少个warp的一个衡量指标**

3.共享内存和寄存器类似，每个SM只有固定大小，所以也影响warp（block

123再结合木桶效应，取最小值就可以得到每个SM能容纳的warp数量

CUDA Occupancy Calculator

Compute Capability:	8.9
Shared Memory Size Config (bytes):	32768
Global Load Cache Mode:	L1+L2 (ca)

setting

Threads Per Block:	32
Registers Per Thread:	1
User Shared Memory Per Block (bytes):	0

input

✓ Apply Automatically · Apply · Reset

Tables · Utilization · Graphs · GPU Data

Occupancy Data:

Property		Value
Active Threads per Multiprocessor	每一个SM中活跃的Threads	768
Active Warps per Multiprocessor	每一个SM中活跃的Warps	24
Active Thread Blocks per Multiprocessor	每一个SM中活跃的Blocks	24
Occupancy of each Multiprocessor	Occupancy	50 %

Allocated Resources:

Resources	Per Block	Limit Per SM	Allocatable Blocks Per SM
Warps (Threads Per Block / Threads Per Warp)	1	48	24
Registers (Warp limit per SM due to per-warp reg count)	1	256	256
Shared Memory (Bytes)	1024	32768	32

Occupancy Limiters:

Physical Limit of GPU (8.9):

Property	Limit
Threads per Warp	
Max Warps per Multiprocessor	
Max Thread Blocks per Multiprocessor	
Max Threads per Multiprocessor	
Maximum Thread Block Size	
Registers per Multiprocessor	
Max Registers per Thread Block	
Max Registers per Thread	
Shared Memory per Multiprocessor (bytes)	
Max Shared Memory per Block	
Register Allocation Unit Size	
Register Allocation Granularity	
Shared Memory Allocation Unit Size	
Warp Allocation Granularity	
Shared Memory Per Block (bytes) (CUDA runtime use)	

Physical Limit of GPU

block_size对SM可启动warp数量的影响

Physical Limit of GPU (8.9):

Property	Limit
Threads per Warp	32
Max Warps per Multiprocessor	48
Max Thread Blocks per Multiprocessor	24
Max Threads per Multiprocessor	1536
Maximum Thread Block Size	1024
Registers per Multiprocessor	65536
Max Registers per Thread Block	65536
Max Registers per Thread	255
Shared Memory per Multiprocessor (bytes)	16384
Max Shared Memory per Block	16384
Register Allocation Unit Size	256
Register Allocation Granularity	warp
Shared Memory Allocation Unit Size	128
Warp Allocation Granularity	4
Shared Memory Per Block (bytes) (CUDA runtime use)	1024

1. 每一个Block必须分配到一个SM中
2. 每一个SM最多可放48个warp(1536个thread)
3. 每一个SM最多可放24个Block

假设需要计算的数据足够多，每个thread都只使用很少的smem和register，设置的Block_Size为(32)

- 根据2可得最多能放48个warp相当于48个Block
- 根据3可得SM最多可放24个Block，所以只能放24个warp
- occupany为0.5=24/48

Allocated Resources:

Resources	Per Block	Limit Per SM	Allocatable Blocks Per SM
Warps (Threads Per Block / Threads Per Warp)	1	48	24
Registers (Warp limit per SM due to per-warp reg count)	1	256	256
Shared Memory (Bytes)	1024	65536	64

block_size对SM可启动warp数量的影响

NVIDIA Nsight Compute

File Connection Debug Profile Tools Window Help

Project Explorer — Default Project

Welcome | test.ncu-rep | Untitled 1 *

Compute Capability:	8.9	Threads Per Block:	32
Shared Memory Size Config (bytes):	65536	Registers Per Thread:	0
Global Load Cache Mode:	L1+L2 (ca)	User Shared Memory Per Block (bytes):	0

✓ Apply Automatically | Apply | Reset

Tables | Graphs | GPU Data

Occupancy Data:

Property	Value
Active Threads per Multiprocessor	768
Active Warps per Multiprocessor	24
Active Thread Blocks per Multiprocessor	24
Occupancy of each Multiprocessor	50 %

Allocated Resources:

Resources	Per Block
Warps (Threads Per Block / Threads Per Warp)	
Registers (Warp limit per SM due to per-warp reg count)	
Shared Memory (Bytes)	

Occupancy Limiters:

Limited By	Blocks per SM
Max Warps or Max Blocks per Multiprocessor	24
Registers per Multiprocessor	24
Shared Memory per Multiprocessor	64

Physical Limit of GPU (8.9):

Property	Li
Threads per Warp	
Max Warps per Multiprocessor	
Max Thread Blocks per Multiprocessor	
Max Threads per Multiprocessor	
Maximum Thread Block Size	
Registers per Multiprocessor	
Max Registers per Thread Block	
Max Registers per Thread	
Shared Memory per Multiprocessor (bytes)	
Max Shared Memory per Block	
Register Allocation Unit Size	
Register Allocation Granularity	
Shared Memory Allocation Unit Size	
Warp Allocation Granularity	
Shared Memory Per Block (bytes) (CUDA runtime use)	

Metric Details

Open an Nsight Compute profiling report, then click a metric on the Details or Raw page. Use the search bar to lookup metrics in the current report.

NVIDIA Nsight Compute

File Connection Debug Profile Tools Window Help

Connect Disconnect Terminate Profile Kernel Baselines Metric Details

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Project Explorer

Search project...

Default Project

Welcome | test.ncu-rep | Untitled 1 *

Compute Capability: 8.9
Shared Memory Size Config (bytes): 65536
Global Load Cache Mode: L1+L2 (ca)
Threads Per Block: 160
Registers Per Thread: 0
User Shared Memory Per Block (bytes): 0

✓ Apply Automatically | Apply | Reset

Tables | Graphs | GPU Data

Occupancy Data:

Property	Value
Active Threads per Multiprocessor	1440
Active Warps per Multiprocessor	45
Active Thread Blocks per Multiprocessor	9
Occupancy of each Multiprocessor	94 %

Physical Limit of GPU (8.9):

Property	Limit
Threads per Warp	32
Max Warps per Multiprocessor	48
Max Thread Blocks per Multiprocessor	24
Max Threads per Multiprocessor	1536
Maximum Thread Block Size	1024
Registers per Multiprocessor	65536
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Shared Memory per Multiprocessor (bytes)	65536
Max Shared Memory per Block	65536
Register Allocation Unit Size	256
Register Allocation Granularity	warp
Shared Memory Allocation Unit Size	128
Warp Allocation Granularity	4
Shared Memory Per Block (bytes) (CUDA runtime use)	1024

Allocated Resources:

esources	Per Block	Limit Per SM	Allocatable Blocks Per SM
hreads Per Warp)	5	48	9
due to per-warp reg count)	5	0	24
	1024	65536	64

Occupancy Limiters:

Limited By	Blocks per SM	Warps Per Block	Warps Per SM
Blocks per Multiprocessor	9	5	45
rocessor	24		
Multiprocessor	64		

Metric Details

Pin Tab | Search metrics in current report

Open an Nsight Compute profiling report, then click a metric on the Details or Raw page.
Use the search bar to lookup metrics in the current report.

最牛逼cuda课程.pptx -...

-4°C 晴朗 搜索 中 拼 8:07 2025/1/4

Register对SM可启动warp数量的影响

Physical Limit of GPU (8.9):

Property	Limit
Threads per Warp	32
Max Warps per Multiprocessor	48
Max Thread Blocks per Multiprocessor	24
Max Threads per Multiprocessor	1536
Maximum Thread Block Size	1024
Registers per Multiprocessor	65536
Max Registers per Thread Block	65536
Max Registers per Thread	255
Shared Memory per Multiprocessor (bytes)	16384
Max Shared Memory per Block	16384
Register Allocation Unit Size	256
Register Allocation Granularity	warp
Shared Memory Allocation Unit Size	128
Warp Allocation Granularity	4
Shared Memory Per Block (bytes) (CUDA runtime use)	1024

1. 每一个SM有65536个寄存器
2. 寄存器的分配单元大小是256个
3. 寄存器的分割粒度是warp
4. warp的分配粒度是4（如果 warp Allocation Granularity 为 4，那么即使一个内核配置只需要 3 个 warp 的寄存器资源，硬件也会按照 4 个 warp 来分配寄存器资源）

假设需要计算的数据足够多，每个线程需要51个register，设置的Block_Size为(128)

- 根据3，每个thread 51 reg，所以每个warp需要51*32=1632个reg
- 根据2，1632个reg，需要按照256粒度分配，1632/256=6.375→7*256=1792，所以每个warp实际需要1792个reg
- 根据1，65536个reg，每个warp需要1792，所以65536/1792=36.57→36个warp
- occupancy=36/48=0.75

Allocated Resources:

Resources	Per Block	Limit Per SM	Allocatable Blocks Per SM
Warps (Threads Per Block / Threads Per Warp)	4	48	12
Registers (Warp limit per SM due to per-warp reg count)	4	36	9
Shared Memory (Bytes)	1024	65536	64

Register对SM可启动warp数量的影响

Physical Limit of GPU (8.9):

Property	Limit
Threads per Warp	32
Max Warps per Multiprocessor	48
Max Thread Blocks per Multiprocessor	24
Max Threads per Multiprocessor	1536
Maximum Thread Block Size	1024
Registers per Multiprocessor	65536
Max Registers per Thread Block	65536
Max Registers per Thread	255
Shared Memory per Multiprocessor (bytes)	16384
Max Shared Memory per Block	16384
Register Allocation Unit Size	256
Register Allocation Granularity	warp
Shared Memory Allocation Unit Size	128
Warp Allocation Granularity	4
Shared Memory Per Block (bytes) (CUDA runtime use)	1024

1. 每一个SM有65536个寄存器
2. 寄存器的分配单元大小是256个
3. 寄存器的分割粒度是warp
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- 根据1，65536个reg，每个warp需要1792，所以65536/1792=36.57→36个warp
- occupancy=36/48=0.75

Allocated Resources:

Resources	Per Block	Limit Per SM	Allocatable Blocks Per SM
Warps (Threads Per Block / Threads Per Warp)	4	48	12
Registers (Warp limit per SM due to per-warp reg count)	4	36	9
Shared Memory (Bytes)	1024	65536	64

Ada GPU Architecture In-Depth

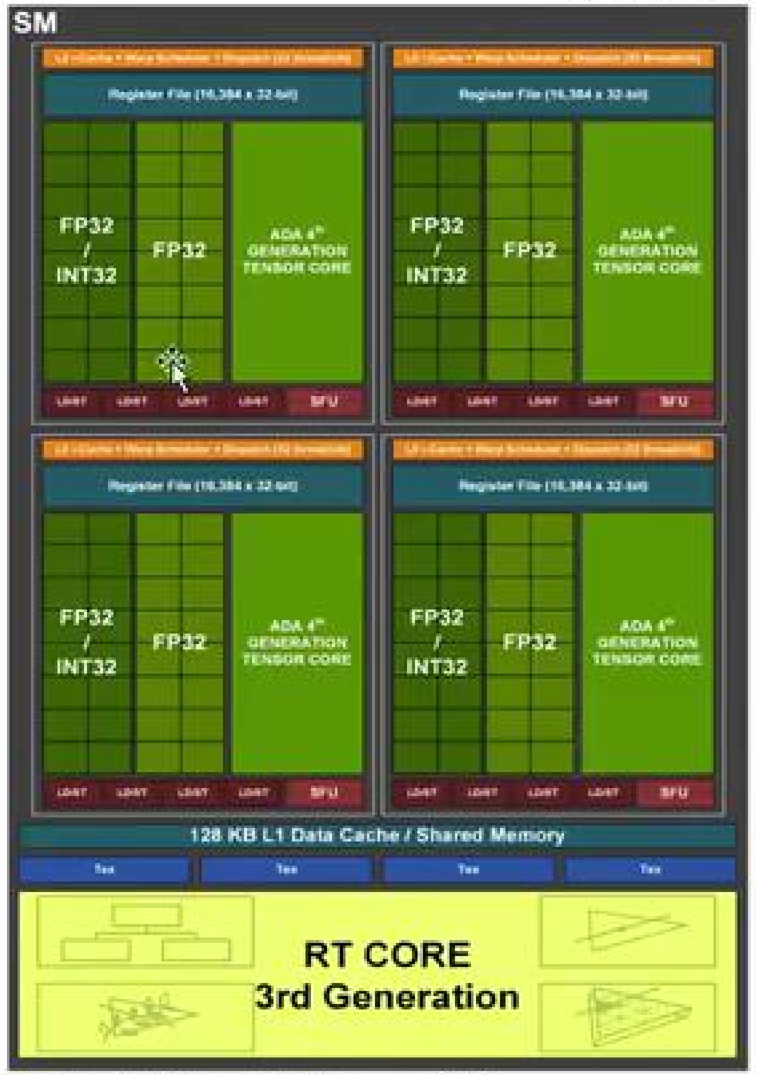

Figure 5. Ada Streaming Multiprocessor (SM)

以左侧图为例

可看出一个SM中有128个Core，所以至少是可以放下128个Thread（4个warp）为了隐藏延时，其实可以放进去更多warp（大于4个）根据显卡的架构不同，这个更多的warp仍然有一个理论上限。

1.用户设置的block的大小会影响实际SM上可以容纳的warp数量

2.CUDA隐藏延迟的手段就是当一个warp处于等待状态的时候会立刻切到下一个warp，而这个立刻就要求线程的切换要特别的迅速（0开销），这与CPU多线程有很大的区别，CPU在进行thread切换的时候，是有开销的（要保留每个thread的上下文），但是GPU不需要因为它有大量的寄存器，每个thread的寄存器数据都是自己用所以就没有切换上下文的需要，自然也可以做到0开销，而与此同时每个SM中的寄存器数量是有上限的所以寄存器也会成为能开启多少个warp的一个衡量指标

3.共享内存和寄存器类似，每个SM只有固定大小，所以也影响warp（block块）的数量。

123再结合木桶效应，取最小值就可以得到每个SM能容纳的warp数量。

block_size对SM可启动warp数量的影响

Physical Limit of GPU (8.9):

Property	Limit
Threads per Warp	32
Max Warps per Multiprocessor	48
Max Thread Blocks per Multiprocessor	24
Max Threads per Multiprocessor	1536
Maximum Thread Block Size	1024
Registers per Multiprocessor	65536
Max Registers per Thread Block	65536
Max Registers per Thread	255
Shared Memory per Multiprocessor (bytes)	16384
Max Shared Memory per Block	16384
Register Allocation Unit Size	256
Register Allocation Granularity	warp
Shared Memory Allocation Unit Size	128
Warp Allocation Granularity	4
Shared Memory Per Block (bytes) (CUDA runtime use)	1024

1. 每一个Block必须分配到一个SM中
2. 每一个SM最多可放48个warp(1536个thread)
3. 每一个SM最多可放24个Block

假设需要计算的数据足够多，每个thread都只使用很少的smem和register，设置的Block_Size为(32)

- 根据2可得最多能放48个warp相当于48个Block
- 根据3可得SM最多可放24个Block，所以只能放24个warp
- occupany为0.5=24/48

Allocated Resources:

Resources	Per Block	Limit Per SM	Allocatable Blocks Per SM
Warps (Threads Per Block / Threads Per Warp)	1	48	24
Registers (Warp limit per SM due to per-warp reg count)	1	256	256
Shared Memory (Bytes)	1024	65536	64

Register对SM可启动warp数量的影响

Physical Limit of GPU (8.9):

Prop
Threads per Warp
Max Warps per Multiprocess
Max Thread Blocks per Mult
Max Threads per Multiproce
Maximum Thread Block Size
Registers per Multiprocesso
Max Registers per Thread Bl
Max Registers per Thread
Shared Memory per Multipro
Max Shared Memory per Blo
Register Allocation Unit Size
Register Allocation Granular
Shared Memory Allocation
Warp Allocation Granularity
Shared Memory Per Block ((

NVIDIA Nsight Compute

File Connection Debug Profile Tools Window Help

Connect Disconnect Terminate Profile Kernel Baselines Metric Details

Project Explorer

Search project...

Default Project

Welcome test.ncu-rep Untitled 1 *

Compute Capability: 8.9 Threads Per Block: 32

Shared Memory Size Config (bytes): 65536 Registers Per Thread: 0

Global Load Cache Mode: L1+L2 (ca) User Shared Memory Per Block (bytes): 0

✓ Apply Automatically Apply Reset

Tables Graphs GPU Data

Occupancy Data:

Property	Value
Active Threads per Multiprocessor	768
Active Warps per Multiprocessor	24
Active Thread Blocks per Multiprocessor	24
Occupancy of each Multiprocessor	50 %

Allocated Resources:

esources	Per Block	Limit Per SM
hreads Per Warp)	1	
due to per-warp reg count)	1	
	1024	

Occupancy Limiters:

Limited By	Blocks per SM	Warps Per Blo
Blocks per Multiprocessor	24	
rocessor	24	
Multiprocessor	64	

Physical Limit of GPU (8.9):

Property	Li
Threads per Warp	
Max Warps per Multiprocessor	
Max Thread Blocks per Multiprocessor	
Max Threads per Multiprocessor	
Maximum Thread Block Size	
Registers per Multiprocessor	
Max Registers per Thread Block	
Max Registers per Thread	
Shared Memory per Multiprocessor (bytes)	
Max Shared Memory per Block	
Register Allocation Unit Size	
Register Allocation Granularity	
Shared Memory Allocation Unit Size	
Warp Allocation Granularity	
Shared Memory Per Block (bytes) (CUDA runtime use)	

Metric Details

Pin Tab Search metrics in current rep

Open an Nsight Compute profiling report, then click a metric on the Details or Raw page. Use the search bar to lookup metrics in the current report.

假设需要

- **根据3，**
- **根据2，**
  reg
- **根据1，**
- **occup**

Register对SM可启动warp数量的影响

Physical Limit of GPU (8.9):

Property	Limit
Threads per Warp	32
Max Warps per Multiprocessor	48
Max Thread Blocks per Multiprocessor	24
Max Threads per Multiprocessor	1536
Maximum Thread Block Size	1024
Registers per Multiprocessor	65536
Max Registers per Thread Block	65536
Max Registers per Thread	255
Shared Memory per Multiprocessor (bytes)	16384
Max Shared Memory per Block	16384
Register Allocation Unit Size	256
Register Allocation Granularity	warp
Shared Memory Allocation Unit Size	128
Warp Allocation Granularity	4
Shared Memory Per Block (bytes) (CUDA runtime use)	1024

1. 每一个SM有65536个寄存器
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3. 寄存器的分割粒度是warp
4. warp的分配粒度是4（如果 warp Allocation Granularity 为 4，那么即使一个内核配置只需要 3 个 warp 的寄存器资源，硬件也会按照 4 个 warp 来分配寄存器资源）

假设需要计算的数据足够多，每个线程需要51

- 根据3，每个thread 51 reg，所以每个wa
- 根据2，1632个reg，需要按照256粒度分 reg
- 根据1，65536个reg，每个warp需要17
- occupancy=36/48=0.75

Allocated Resources:

Resources
Warps (Threads Per Block / Threads Per Warp)
Registers (Warp limit per SM due to per-warp reg count)
Shared Memory (Bytes)

NVIDIA Nsight Compute

File Connection Debug Profile Tools Window Help

Project Explorer

Default Project

Compute Capability: 8.9
Threads Per Block: 128
Shared Memory Size Config (bytes): 65536
Registers Per Thread: 0
Global Load Cache Mode:
User Shared Memory Per Block (bytes): 0

Apply Automatically Reset

Tables GPU Data

Occupancy Data

Property	Value
Active Threads per Multiprocessor	1536
Active Warps per Multiprocessor	48
Active Thread Blocks per Multiprocessor	12
Occupancy of each Multiprocessor	100 %

Allocated Resources:

Resources	Per Block	Limit Per SM
Threads Per Warp)	4	
due to per-warp reg count)	4	

Physical Limit of GPU (8.9):

Property
Threads per Warp
Max Warps per Multiprocessor
Max Thread Blocks per Multiprocessor
Max Threads per Multiprocessor
Maximum Thread Block Size
Registers per Multiprocessor
Max Registers per Thread Block
Max Registers per Thread

Register对SM可启动warp数量的影响

Physical Limit of GPU (8.9):

Property
Threads per Warp
Max Warps per Multiprocessor
Max Thread Blocks per Multiprocessor
Max Threads per Multiprocessor
Maximum Thread Block Size
Registers per Multiprocessor
Max Registers per Thread Block
Max Registers per Thread
Shared Memory per Multiprocessor (bytes)
Max Shared Memory per Block
Register Allocation Unit Size
Register Allocation Granularity
Shared Memory Allocation Unit Size
Warp Allocation Granularity
Shared Memory Per Block (bytes) (CUDA runtime use)

假设需要计算的数

- 根据3，每个thr
- 根据2，1632个 reg
- 根据1，65536
- occupancy=3

NVIDIA Nsight Compute

File Connection Debug Profile Tools Window Help

Connect Disconnect Terminate Profile Kernel Baselines Metric Details

Project Explorer

Default Project

Welcome test.ncu-rep Untitled 1 *

Compute Capability: 8.9 Threads Per Block: 128

Shared Memory Size Config (bytes): 65536 Registers Per Thread: 0

Global Load Cache Mode: L1+L2 (ca) User Shared Memory Per Block (bytes): 0

Apply Automatically Apply Reset

Tables Graphs GPU Data

Occupancy Data:

Property	Value
Active Threads per Multiprocessor	1536
Active Warps per Multiprocessor	48
Active Thread Blocks per Multiprocessor	12
Occupancy of each Multiprocessor	100 %

Allocated Resources:

esources	Per Block	Limit Per SM
hreads Per Warp)	4	
due to per-warp reg count)	4	
	1024	

Occupancy Limiters:

Limited By	Blocks per SM	Warps Per Blo
Blocks per Multiprocessor	12	
rocessor	24	
Multiprocessor	64	

Physical Limit of GPU (8.9):

Property	Li
Threads per Warp	
Max Warps per Multiprocessor	
Max Thread Blocks per Multiprocessor	
Max Threads per Multiprocessor	
Maximum Thread Block Size	
Registers per Multiprocessor	
Max Registers per Thread Block	
Max Registers per Thread	
Shared Memory per Multiprocessor (bytes)	
Max Shared Memory per Block	
Register Allocation Unit Size	
Register Allocation Granularity	
Shared Memory Allocation Unit Size	
Warp Allocation Granularity	
Shared Memory Per Block (bytes) (CUDA runtime use)	

Metric Details

Open an Nsight Compute profiling re then click a metric on the Details or page. Use the search bar to lookup metric the current report.

Allocated Reso

Warps (Threa

Registers (Wa

Shared Mem

Register对SM可启动warp数量的影响

Physical Limits of GPU (8.9)

Property	Limit
Threads per Warp	32
Max Warps per Multiprocessor	48
Max Thread Blocks per Multiprocessor	24
Max Threads per Multiprocessor	1536
Maximum Thread Block Size	1024
Registers per Multiprocessor	65536
Max Registers per Thread Block	65536
Max Registers per Thread	255
Shared Memory per Multiprocessor (bytes)	16384
Max Shared Memory per Block	16384
Register Allocation Unit Size	256
Register Allocation Granularity	warp
Shared Memory Allocation Unit Size	128
Warp Allocation Granularity	4
Shared Memory Per Block (bytes) (CUDA runtime use)	1024

1. 每一个SM有65536个寄存器
2. 寄存器的分配单元大小是256个
3. 寄存器的分割粒度是warp
4. warp的分配粒度是4（如果 warp Allocation Granularity 为 4，那么即使一个内核配置只需要 3 个 warp 的寄存器资源，硬件也会按照 4 个 warp 来分配寄存器资源）

假设需要计算的数据足够多，每个线程需要51个register，设置的Block_Size为(128)

- 根据3，每个thread 51 reg，所以每个warp需要51*32=1632个reg
- 根据2，1632个reg，需要按照256粒度分配，1632/256=6.375→7*256=1792，所以每个warp实际需要1792个reg
- 根据1，65536个reg，每个warp需要1792，所以65536/1792=36.57→36个warp
- occupancy=36/48=0.75

Allocated Resources:

Resources	Per Block	Limit Per SM	Allocatable Blocks Per SM
Warps (Threads Per Block / Threads Per Warp)	4	48	12
Registers (Warp limit per SM due to per-warp reg count)	4	36	9
Shared Memory (Bytes)	1024	65536	64

Register对SM可启动warp数量的影响

Physical Limit of GPU (8.9)

Property	Limit
Threads per Warp	32
Max Warps per Multiprocessor	48
Max Thread Blocks per Multiprocessor	24
Max Threads per Multiprocessor	1536
Maximum Thread Block Size	1024
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假设需要计算的数据足够多，每个线程需要90个register，设置的Block_Size为(128)

- 根据3，每个thread 90 reg，所以每个warp需要90*32=2880个reg
- 根据2，2880个reg，需要按照256粒度分配，2880/256=11.25→12*256=3072，所以每个warp实际需要3072个reg
- 根据1，65536个reg，每个warp需要3072，所以65536/3072=21.3333→21个warp
- 根据4，需要按照4个warp来分配，但是分配24个reg资源不够，所以只能分配20个warp
- occupanc

Allocated Resources:

Resources	Per Block	Limit Per SM	Allocatable Blocks Per SM
Warps (Threads Per Block / Threads Per Warp)	4	48	12
Registers (Warp limit per SM due to per-warp reg count)	4	20	5
Shared Memory (Bytes)	1024	65536	64

Register对SM可启动warp数量的影响

假设需要计算的数

- 根据3，每个t
- 根据2，2880
reg
- 根据1，6553
- 根据4，需要扌
- occupanc

NVIDIA Nsight Compute

File Connection Debug Profile Tools Window Help

Project Explorer | Welcome | test.ncu-rep | Untitled 1 *

Default Project

Compute Capability:	8.9	Threads Per Block:	128
Shared Memory Size Config (bytes):	65536	Registers Per Thread:	51
Global Load Cache Mode:	L1+L2 (ca)	User Shared Memory Per Block (bytes):	0

✓ Apply Automatically　Apply　Reset

Tables　Graphs　GPU Data

Occupancy Data:

Property	Value
Active Threads per Multiprocessor	1152
Active Warps per Multiprocessor	36
Active Thread Blocks per Multiprocessor	9
Occupancy of each Multiprocessor	75 %

Allocated Resources:

esources	Per Block	Limit Per SM
hreads Per Warp)	4	
due to per-warp reg count)	4	
	1024	

Occupancy Limiters:

Limited By	Blocks per SM	Warps Per Blo
Blocks per Multiprocessor	12	
rocessor	9	
Multiprocessor	64	

Physical Limit of GPU (8.9):

Property	Li
Threads per Warp	
Max Warps per Multiprocessor	
Max Thread Blocks per Multiprocessor	
Max Threads per Multiprocessor	
Maximum Thread Block Size	
Registers per Multiprocessor	
Max Registers per Thread Block	
Max Registers per Thread	
Shared Memory per Multiprocessor (bytes)	
Max Shared Memory per Block	
Register Allocation Unit Size	
Register Allocation Granularity	
Shared Memory Allocation Unit Size	
Warp Allocation Granularity	
Shared Memory Per Block (bytes) (CUDA runtime use)	

Metric Details

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Register对SM可启动warp数量的影响

Physical Limit of GPU (8.9)

Property	Limit
Threads per Warp	32
Max Warps per Multiprocessor	48
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Max Threads per Multiprocessor	1536
Maximum Thread Block Size	1024
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- 根据1，65536个reg，每个warp需要3072，所以65536/3072=21.3333→21个warp
- 根据4，需要按照4个warp来分配，但是分配24个reg资源不够，所以只能分配20个warp
- occupancy=20/48=0.41666

Allocated Resources:

Resources	Per Block	Limit Per SM	Allocatable Blocks Per SM
Warps (Threads Per Block / Threads Per Warp)	4	48	12
Registers (Warp limit per SM due to per-warp reg count)	4	20	5
Shared Memory (Bytes)	1024	65536	64

Register对SM可启动warp数量的影响

Physical Limit of GPU (8.9)

Property	Limit
Threads per Warp	32
Max Warps per Multiprocessor	48
Max Thread Blocks per Multiprocessor	24
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Shared Memory Per Block (bytes) (CUDA runtime use)	1024

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假设需要计算的数据足够多，每个线程需要90个register，设置的Block_Size为(128)

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- 根据4，需要按照4个warp来分配，但是分配24个reg资源不够，所以只能分配20个warp
- occupancy=20/48=0.41666

Allocated Resources:

Resources	Per Block	Limit Per SM	Allocatable Blocks Per SM
Warps (Threads Per Block / Threads Per Warp)	4	48	12
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Shared Memory (Bytes)	1024	65536	64

Register对SM可启动warp数量的影响

Physical Limit of GPU (8.9)

Property	Limit
Threads per Warp	32
Max Warps per Multiprocessor	48
Max Thread Blocks per Multiprocessor	24
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假设需要计算的数据足够多，每个线程需要90个register，设置的Block_Size为(128)

- 根据3，每个thread 90 reg，所以每个warp需要90*32=2880个reg
- 根据2，2880个reg，需要按照256粒度分配，2880/256=11.25→12*256=3072，所以每个warp实际需要3072个reg
- 根据1，65536个reg，每个warp需要3072，所以65536/3072=21.3333→21个warp
- 根据4，需要按照4个warp来分配，但是分配24个reg资源不够，所以只能分配20个warp
- occupancy=20/48=0.41666

Allocated Resources:

Resources	Per Block	Limit Per SM	Allocatable Blocks Per SM
Warps (Threads Per Block / Threads Per Warp)	4	48	12
Registers (Warp limit per SM due to per-warp reg count)	4	20	5
Shared Memory (Bytes)	1024	65536	64

smem对SM可启动warp数量的影响

Physical Limit of GPU (8.9):

Property	Limit
Threads per Warp	32
Max Warps per Multiprocessor	48
Max Thread Blocks per Multiprocessor	24
Max Threads per Multiprocessor	1536
Maximum Thread Block Size	1024
Registers per Multiprocessor	65536
Max Registers per Thread Block	65536
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Register Allocation Unit Size	256
Register Allocation Granularity	warp
Shared Memory Allocation Unit Size	128
Warp Allocation Granularity	4
Shared Memory Per Block (bytes) (CUDA runtime use)	1024

1. 每一个SM有16384Byte smem
2. smem的分配单元大小是128Byte
3. 每个SM都必须分配1024Byte smem
4. smem配置的大小为32768Byte

Compute Capability:	8.9	Threads Per Block:	128
Shared Memory Size Config (bytes):	32768	Registers Per Thread:	1
Global Load Cache Mode:		User Shared Memory Per Block (bytes):	5000

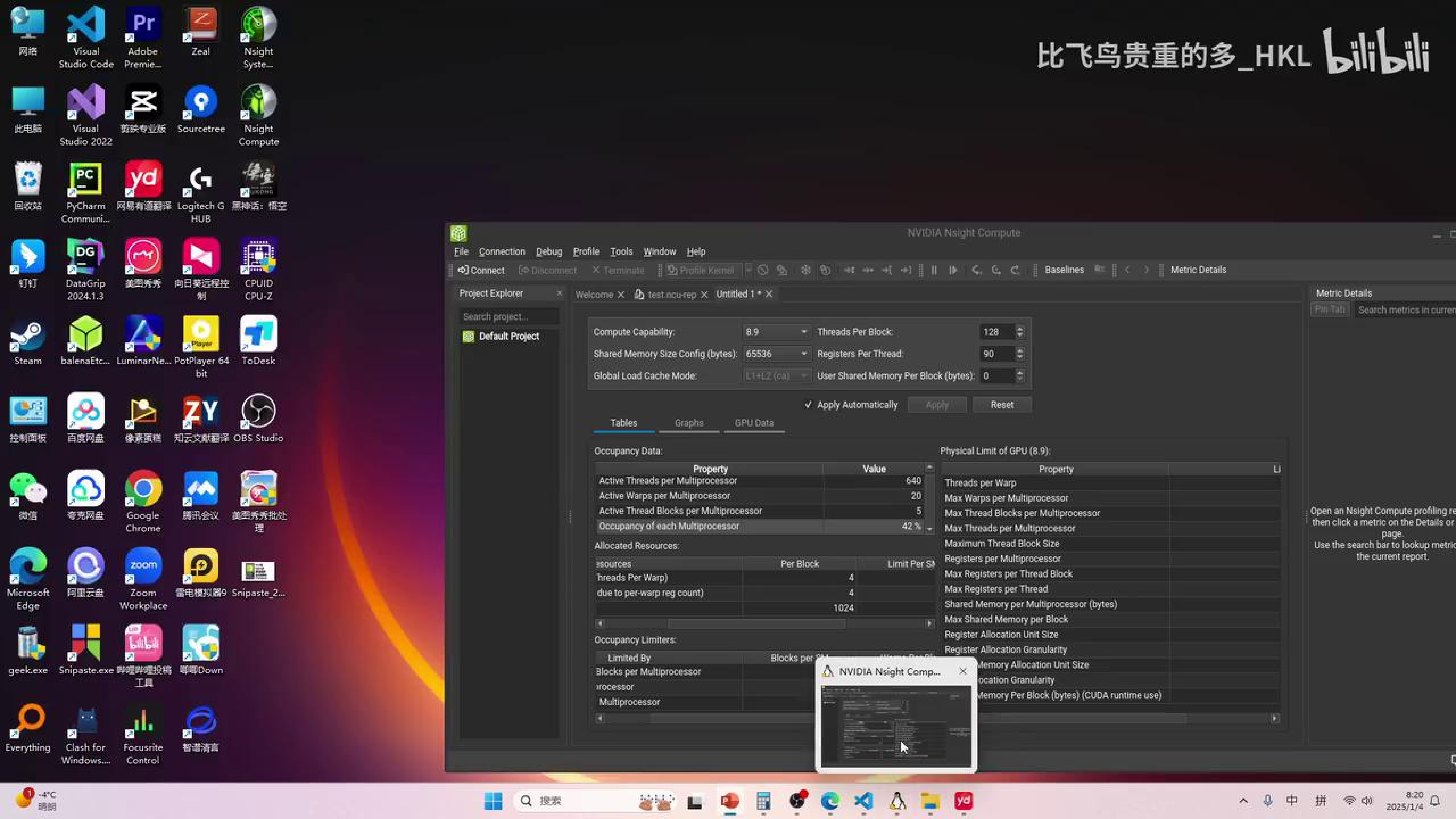
✓ Apply Automatically Apply Reset

假设需要计算的数据足够多，每个线程需要5000Byte smem，设置的Block_Size为(128)

- 根据3，每个Block需要1024+5000=6024Byte smem
- 根据2，6024Byte，需要按照128Byte粒度分配，6024/128=47.06→48*128=6144，所以每个Block实际需要6144Byte
- 根据4，一共32768Byte，32768/6144=5.33，所以只能分配5个Block
- occupancy=(5*4)/48=0.41666

Allocated Resources:

Resources	Per Block	Limit Per SM	Allocatable Blocks Per SM
Warps (Threads Per Block / Threads Per Warp)	4	48	12
Registers (Warp limit per SM due to per-warp reg count)	4	256	64
Shared Memory (Bytes)	6144	32768	5



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NVIDIA Nsight Compute
File Connection Debug Profile Tools Window Help
Connect Disconnect Terminate Profile Kernel Baselines Metric Details
Project Explorer
Search project...
Default Project
Welcome test.ncu-rep Untitled 1 *
Compute Capability: 8.9
Threads Per Block: 128
Shared Memory Size Config (bytes): 65536
Registers Per Thread: 90
Global Load Cache Mode: L1+L2 (ca)
User Shared Memory Per Block (bytes): 0
Apply Automatically Apply Reset
Tables Graphs GPU Data
Occupancy Data:
Property Value
Active Threads per Multiprocessor 640
Active Warps per Multiprocessor 20
Active Thread Blocks per Multiprocessor 5
Occupancy of each Multiprocessor 42 %
Allocated Resources:
Per Block
Limit Per SM
due to per-warp reg count) 4
1024
Occupancy Limiters:
Limited By
Blocks per Multiprocessor
Multiprocessor
Physical Limit of GPU (8.9):
Property
Threads per Warp
Max Warps per Multiprocessor
Max Thread Blocks per Multiprocessor
Max Threads per Multiprocessor
Maximum Thread Block Size
Registers per Multiprocessor
Max Registers per Thread Block
Max Registers per Thread
Shared Memory per Multiprocessor (bytes)
Max Shared Memory per Block
Register Allocation Unit Size
Register Allocation Granularity
Metric Details
NVIDIA Nsight Comp...

smem对SM可启动warp数量的影响

Physical Limit of GPU (8.9):

Property	Limit
Threads per Warp	32
Max Warps per Multiprocessor	48
Max Thread Blocks per Multiprocessor	24
Max Threads per Multiprocessor	1536
Maximum Thread Block Size	1024
Registers per Multiprocessor	65536
Max Registers per Thread Block	65536
Max Registers per Thread	255
Shared Memory per Multiprocessor (bytes)	16384
Max Shared Memory per Block	16384
Register Allocation Unit Size	256
Register Allocation Granularity	warp
Shared Memory Allocation Unit Size	128
Warp Allocation Granularity	4
Shared Memory Per Block (bytes) (CUDA runtime use)	1024

1. 每一个SM有16384Byte smem
2. smem的分配单元大小是128Byte
3. 每个SM都必须分配1024Byte smem
4. smem配置的大小为32768Byte

Compute Capability:	8.9	Threads Per Block:	128
Shared Memory Size Config (bytes):	32768	Registers Per Thread:	1
Global Load Cache Mode:		User Shared Memory Per Block (bytes):	5000

✓ Apply Automatically

假设需要计算的数据足够多，每个线程需要5000Byte smem，设置的Block_Size为(128)

- 根据3，每个Block需要1024+5000=6024Byte smem
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smem对SM可启动warp数量的影响

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Max Threads per Multiprocessor	1536
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Max Registers per Thread Block	65536
Max Registers per Thread	255
Shared Memory per Multiprocessor (bytes)	16384
Max Shared Memory per Block	16384
Register Allocation Unit Size	256
Register Allocation Granularity	warp
Shared Memory Allocation Unit Size	128
Warp Allocation Granularity	4
Shared Memory Per Block (bytes) (CUDA runtime use)	1024

1.
2.
3.
4.

假设需要计算的数据足够多，每个线程需要5000By

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- occupancy=(5*4)/48=0.41666

Allocated Resources:

Resources
Warps (Threads Per Block / Threads Per Warp)
Registers (Warp limit per SM due to per-warp reg count)
Shared Memory (Bytes)

计算器

标准

20 ÷ 48 =

0.4166666666666667

NVIDIA Nsight Compute

File Connection Debug Profile Tools Window Help

Project Explorer — Default Project

Welcome | test.ncu-rep | Untitled 1 *

Compute Capability: 8.9 | Threads Per Block: 128

Shared Memory Size Config (bytes): 32768 | Registers Per Thread: 90

Global Load Cache Mode: L1+L2 (ca) | User Shared Memory Per Block (bytes): 0

✓ Apply Automatically | Apply | Reset

Tables | Graphs | GPU Data

Occupancy Data:

Property	Value
Active Threads per Multiprocessor	640
Active Warps per Multiprocessor	20
Active Thread Blocks per Multiprocessor	5
Occupancy of each Multiprocessor	42 %

Allocated Resources:

esources	Per Block	Limit Per SM
hreads Per Warp)	4	
due to per-warp reg count)	4	
	1024	

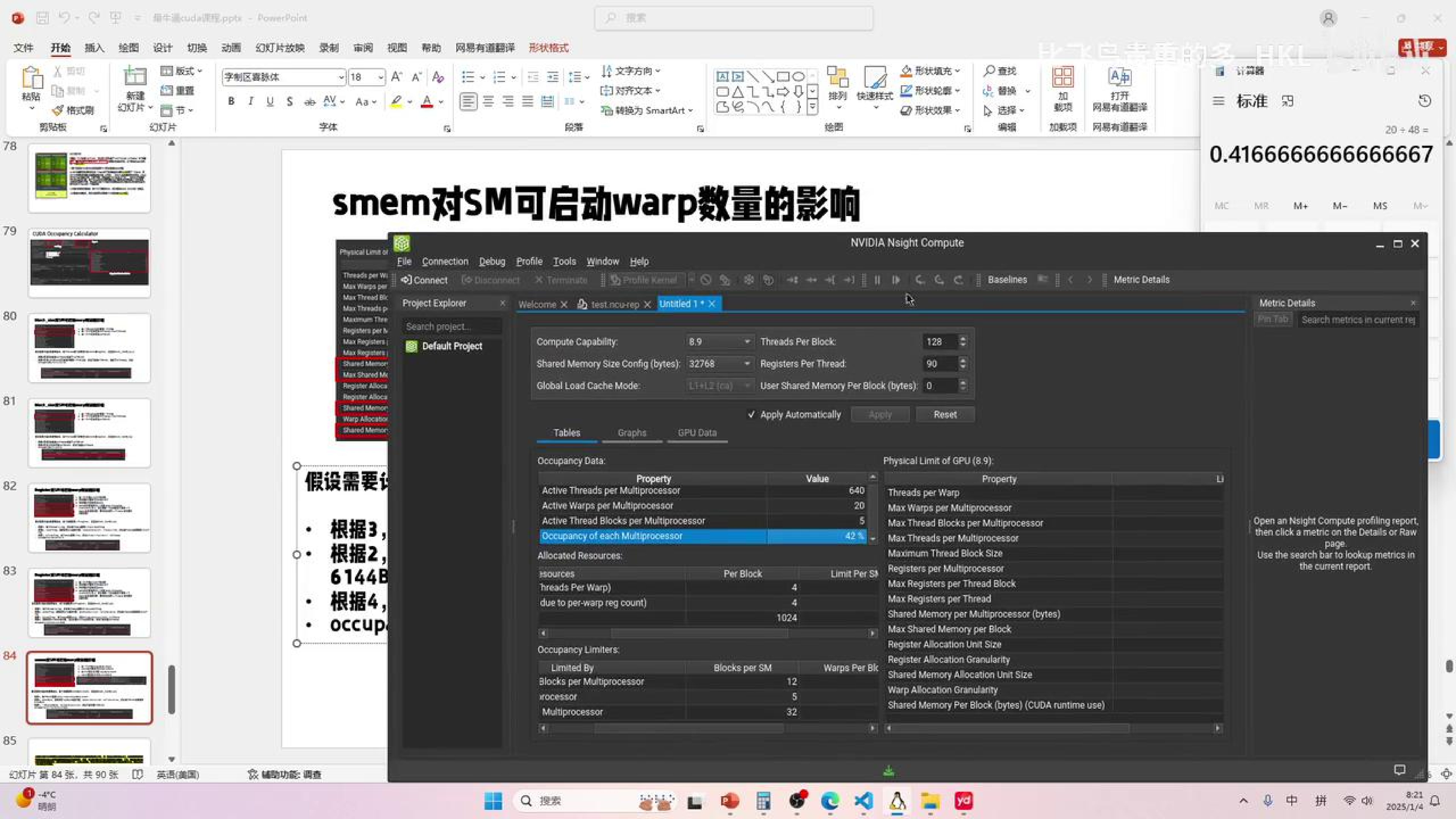
Occupancy Limiters:

Limited By	Blocks per SM	Warps Per Blk
Blocks per Multiprocessor	12	
rocessor	5	
Multiprocessor	32	

Physical Limit of GPU (8.9):

Property
Threads per Warp
Max Warps per Multiprocessor
Max Thread Blocks per Multiprocessor
Max Threads per Multiprocessor
Maximum Thread Block Size
Registers per Multiprocessor
Max Registers per Thread Block
Max Registers per Thread
Shared Memory per Multiprocessor (bytes)
Max Shared Memory per Block
Register Allocation Unit Size
Register Allocation Granularity
Shared Memory Allocation Unit Size
Warp Allocation Granularity
Shared Memory Per Block (bytes) (CUDA runtime use)

幻灯片 第 84 张，共 90 张



smem对SM可启动warp数量的影响
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根据3，
根据2，
6144B
根据4，
occupa
NVIDIA Nsight Compute
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hreads Per Warp) 4
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Multiprocessor 32
Physical Limit of GPU (8.9):
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Max Warps per Multiprocessor
Max Thread Blocks per Multiprocessor
Max Threads per Multiprocessor
Maximum Thread Block Size
Registers per Multiprocessor
Max Registers per Thread Block
Max Registers per Thread
Shared Memory per Multiprocessor (bytes)
Max Shared Memory per Block
Register Allocation Unit Size
Register Allocation Granularity
Shared Memory Allocation Unit Size
Warp Allocation Granularity
Shared Memory Per Block (bytes) (CUDA runtime use)
Metric Details
Open an Nsight Compute profiling report, then click a metric on the Details or Raw page.
Use the search bar to lookup metrics in the current report.
计算器
标准
20 ÷ 48 =
0.4166666666666667
幻灯片 第 84 张，共 90 张

占用率是指每个多处理器上活跃线程束的数量与可能的最大活跃线程束数量之比。另一种看待占用率的方式是，将其视为硬件处理线程束的能力中正在被积极使用的那部分所占的百分比。然而，较高的占用率并不总是能带来更高的性能，不过，较低的占用率总会降低隐藏延迟的能力，进而导致整体性能下降。在执行过程中，理论占用率和实际达到的占用率之间存在较大差异，通常意味着工作负载高度不均衡。

Occupancy

Occupancy is the ratio of the number of active warps per multiprocessor to the maximum number of possible active warps. Another way to view occupancy is the percentage of the hardware's ability to process warps that is actively in use. Higher occupancy does not always result in higher performance, however, low occupancy always reduces the ability to hide latencies, resulting in overall performance degradation. Large discrepancies between the theoretical and the achieved occupancy during execution typically indicates highly imbalanced workloads.

Theoretical Occupancy [%]	100	Block Limit Registers [block]	64
Theoretical Active Warps per SM [warp]	48	Block Limit Shared Mem [block]	32
Achieved Occupancy [%]	86.57	Block Limit Warps [block]	24
Achieved Active Warps Per SM [warp]	41.55	Block Limit SM [block]	24

Achieved Occupancy
Est. Speedup: 6.03%

The difference between calculated theoretical (100.0%) and measured achieved occupancy (86.6%) can be the result of warp scheduling overheads or workload imbalances during the kernel execution. Load imbalances can occur between warps within a block as well as across blocks of the same kernel. See the CUDA Best Practices Guide for more details on optimizing occupancy.

NVIDIA Nsight Compute

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Project Explorer

Search project...

Default Project

Welcome | test.ncu-rep | Untitled 1 *

Page: Details Result: 0 - 529 - transpose_naive Add Baseline Apply Rules Occupancy Calculator Copy as Image

	Result	Time	Cycles	Regs	GPU	SM Frequency	CC	Process
Current	529 - transpose_naive (64, 64, 1)x(8, 32, 1)	9.57 usecond	21,596	16	0 - NVIDIA GeForce RTX 4090	2.25 cycle/nsecond	8.9	[544] my_transpose

Launch Statistics

Summary of the configuration used to launch the kernel. The launch configuration defines the size of the kernel grid, the division of the grid into blocks, and the GPU resources needed to execute the kernel. Choosing an efficient launch configuration maximizes device utilization.

Grid Size	4096	Function Cache Configuration	CachePreferNone
Registers Per Thread [register/thread]	16	Static Shared Memory Per Block [byte/block]	0
Block Size	256	Dynamic Shared Memory Per Block [byte/block]	0
Threads [thread]	1048576	Driver Shared Memory Per Block [Kbyte/block]	1.02
Waves Per SM	5.33	Shared Memory Configuration Size [Kbyte]	16.38

Occupancy

Occupancy is the ratio of the number of active warps per multiprocessor to the maximum number of possible active warps. Another way to view occupancy is the percentage of the hardware's ability to process warps that is actively in use. Higher occupancy does not always result in higher performance, however, low occupancy always reduces the ability to hide latencies, resulting in overall performance degradation. Large discrepancies between the theoretical and the achieved occupancy during execution typically indicates highly imbalanced workloads.

Theoretical Occupancy [%]	100	Block Limit Registers [block]	16
Theoretical Active Warps per SM [warp]	48	Block Limit Shared Mem [block]	16
Achieved Occupancy [%]	72.60	Block Limit Warps [block]	6
Achieved Active Warps Per SM [warp]	34.85	Block Limit SM [block]	24

Occupancy Limiters
Est. Speedup: 27.40%

This kernel's theoretical occupancy is not impacted by any block limit. The difference between calculated theoretical (100.0%) and measured achieved occupancy (72.6%) can be the result of warp scheduling overheads or workload imbalances during the kernel execution. Load imbalances can occur between warps within a block as well as across blocks of the same kernel. See the CUDA Best Practices Guide for more details on optimizing occupancy.

Source Counters

Source metrics, including branch efficiency and sampled warp stall reasons. Warp Stall Sampling metrics are periodically sampled over the kernel runtime. They indicate when warps were stalled and couldn't be scheduled. See the documentation for a description of all stall reasons. Only focus on stalls if the schedulers fail to issue every cycle.

Branch Instructions [inst]	32768	Branch Efficiency [%]	0
Branch Instructions Ratio [%]	0.06	Avg. Divergent Branches	0

Uncoalesced Global Accesses

This kernel has uncoalesced global accesses resulting in a total of 131072 excessive sectors (33% of the total 393216 sectors). Check the L2 Theoretical Sectors Global Excessive table for the primary source locations. The CUDA Programming Guide had additional information on reducing uncoalesced device memory accesses.

L2 Theoretical Sectors Global Excessive

Location	Value	Value (%)
my_transpose.cu:50 (0x700e58be0 in transpose_naive)	131,072	100

Follow the *rules outputs* to get guidance on how to navigate through the report and quickly discover performance bottlenecks in this kernel.
You could also disable individual sections to focus on selected performance aspects and make profiling faster.

Metric Details

Pin Tab Search metrics in current report

Open an Nsight Compute profiling report, then click a metric on the Details or Raw page.
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NVIDIA Nsight Compute

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Project Explorer

Search project...

Default Project

Welcome | test.ncu-rep | Untitled 1 *

Page: Details Result: 3 - 559 - transpose_naive Add Baseline Apply Rules Occupancy Calculator Copy as Image

	Result	Time	Cycles	Regs	GPU	SM Frequency	CC	Process
Current	559 - transpose_naive (64, 64, 1)x(8, 32, 1)	8.80 usecond	19,529	16	0 - NVIDIA GeForce RTX 4090	2.21 cycle/nsecond	8.9	[544] my_transpose

Executed Ipc Elapsed [inst/cycle]	0.21	SM Busy [%]	5.86
Executed Ipc Active [inst/cycle]	0.22	Issue Slots Busy [%]	5.86
Issued Ipc Active [inst/cycle]	0.23		

Low Utilization All compute pipelines are under-utilized. Either this kernel is very small or it doesn't issue enough warps per scheduler. Check the Launch Statistics and Scheduler Statistics sections for further details.

Memory Workload Analysis

Detailed analysis of the memory resources of the GPU. Memory can become a limiting factor for the overall kernel performance when fully utilizing the involved hardware units (Mem Busy), exhausting the available communication bandwidth between those units (Max Bandwidth), or by reaching the maximum throughput of issuing memory instructions (Mem Pipes Busy). Detailed chart of the memory units. Detailed tables with data for each memory unit.

Memory Throughput [Gbyte/second]	476.64	Mem Busy [%]	43.79
L1/TEX Hit Rate [%]	52.01	Max Bandwidth [%]	74.54
L2 Hit Rate [%]	62.97	Mem Pipes Busy [%]	10.51
L2 Compression Success Rate [%]	0	L2 Compression Ratio	0

L1TEX Global Store Access Pattern
Est. Speedup: 0.49%
The memory access pattern for global stores in L1TEX might not be optimal. On average, this kernel accesses 4.0 bytes per thread per memory request; but the address pattern, possibly caused by the stride between threads, results in 8.0 sectors per request, or 8.0*32 = 256.0 bytes of cache data transfers per request. The optimal thread address pattern for 4.0 byte accesses would result in 4.0*32 = 128.0 bytes of cache data transfers per request, to maximize L1TEX cache performance. Check the Source Counters section for uncoalesced global stores.

L2 Load Access Pattern
Est. Speedup: 7.73%
The memory access pattern for loads from L1TEX to L2 is not optimal. The granularity of an L1TEX request to L2 is a 128 byte cache line. That is 4 consecutive 32-byte sectors per L2 request. However, this kernel only accesses an average of 1.1 sectors out of the possible 4 sectors per cache line. Check the Source Counters section for uncoalesced loads and try to minimize how many cache lines need to be accessed per memory request.

L2 Store Access Pattern
Est. Speedup: 22.33%
The memory access pattern for stores from L1TEX to L2 is not optimal. The granularity of an L1TEX request to L2 is a 128 byte cache line. That is 4 consecutive 32-byte sectors per L2 request. However, this kernel only accesses an average of 1.4 sectors out of the possible 4 sectors per cache line. Check the Source Counters section for uncoalesced stores and try to minimize how many cache lines need to be accessed per memory request.

Scheduler Statistics

Summary of the activity of the schedulers issuing instructions. Each scheduler maintains a pool of warps that it can issue instructions for. The upper bound of warps in the pool (Theoretical Warps) is limited by the launch configuration. On every cycle each scheduler checks the state of the allocated warps in the pool (Active Warps). Active warps that are not stalled (Eligible Warps) are ready to issue their next instruction. From the set of eligible warps the scheduler selects a single warp from which to issue one or more instructions (Issued Warp). On cycles with no eligible warps, the issue slot is skipped and no instruction is issued. Having many skipped issue slots indicates poor latency hiding.

Active Warps Per Scheduler [warp]	8.77	No Eligible [%]	92.88
Eligible Warps Per Scheduler [warp]	0.11	One or More Eligible [%]	7.12
Issued Warp Per Scheduler	0.07		

Issue Slot Utilization
Est. Speedup: 92.88%
Every scheduler is capable of issuing one instruction per cycle, but for this kernel each scheduler only issues an instruction every 14.1 cycles. This might leave hardware resources underutilized and may lead to less optimal performance. Out of the maximum of 12 warps per scheduler, this kernel allocates an average of 8.77 active warps per scheduler, but only an average of 0.11 warps were eligible per cycle. Eligible warps are the subset of active warps that are ready to issue their next instruction. Every cycle with no eligible warp results in no instruction being issued and the issue slot remains unused. To increase the number of eligible warps, reduce the time the active warps are stalled by inspecting the top stall reasons on the Warp State Statistics and Source Counters sections.

Metric Details

Search metrics in current report

Open an Nsight Compute profiling report, then click a metric on the Details or Raw page.
Use the search bar to lookup metrics in the current report.

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Project Explorer

Search project...

Default Project

Welcome | test.ncu-rep | Untitled 1 *

Page: Details | Result: 3 - 559 - transpose_naive | Add Baseline | Apply Rules | Occupancy Calculator | Copy as Image

	Result	Time	Cycles	Regs	GPU	SM Frequency	CC	Process
Current	559 - transpose_naive (64, 64, 1)x(8, 32, 1)	8.80 usecond	19,529	16	0 - NVIDIA GeForce RTX 4090	2.21 cycle/nsecond	8.9	[544] my_transpose

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Warp State Statistics

Analysis of the states in which all warps spent cycles during the kernel execution. The warp states describe a warp's readiness or inability to issue its next instruction. The warp cycles per instruction define the latency between two consecutive instructions. The higher the value, the more warp parallelism is required to hide this latency. For each warp state, the chart shows the average number of cycles spent in that state per issued instruction. Stalls are not always impacting the overall performance nor are they completely avoidable. Only focus on stall reasons if the schedulers fail to issue every cycle. When executing a kernel with mixed library and user code, these metrics show the combined values.

Metric Details

Pin Tab | Search metrics in current report

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Project Explorer

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Default Project

Welcome | test.ncu-rep | Untitled 1 *

Page: Details | Result: 3 - 559 - transpose_naive | Add Baseline | Apply Rules | Occupancy Calculator | Copy as Image

	Result	Time	Cycles	Regs	GPU	SM Frequency	CC	Process
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Every scheduler is capable of issuing one instruction per cycle, but for this kernel each scheduler only issues an instruction every 14.1 cycles. This might leave hardware resources underutilized and may lead to less optimal performance. Out of the maximum of 12 warps per scheduler, this kernel allocates an average of 8.77 active warps per scheduler, but only an average of 0.11 warps were eligible per cycle. Eligible warps are the subset of active warps that are ready to issue their next instruction. Every cycle with no eligible warp results in no instruction being issued and the issue slot remains unused. To increase the number of eligible warps, reduce the time the active warps are stalled by inspecting the top stall reasons on the Warp State Statistics and Source Counters sections.

Warp State Statistics

Analysis of the states in which all warps spent cycles during the kernel execution. The warp states describe a warp's readiness or inability to issue its next instruction. The warp cycles per instruction define the latency between two consecutive instructions. The higher the value, the more warp parallelism is required to hide this latency. For each warp state, the chart shows the average number of cycles spent in that state per issued instruction. Stalls are not always impacting the overall performance nor are they completely avoidable. Only focus on stall reasons if the schedulers fail to issue every cycle. When executing a kernel with mixed library and user code, these metrics show the combined values.

Warp Cycles Per Issued Instruction [cycle]	123.25	Avg. Active Threads Per Warp	32
Warp Cycles Per Executed Instruction [cycle]	130.08	Avg. Not Predicated Off Threads Per Warp	32

Long Scoreboard
Est. Speedup: 84.75%

On average, each warp of this kernel spends 104.5 cycles being stalled waiting for a scoreboard dependency on a L1TEX (local, global, surface, texture) operation. Find the instruction producing the data being waited upon to identify the culprit. To reduce the number of cycles waiting on L1TEX data accesses verify the memory access patterns are optimal for the target architecture, attempt to increase cache hit rates by increasing data locality (coalescing), or by changing the cache configuration. Consider moving frequently used data to shared memory. This stall type represents about 84.7% of the total average of 123.2 cycles between issuing two instructions.

Warp Stall Check the Warp Stall Sampling (All Samples) table for the top stall locations in your source based on sampling data. The Kernel Profiling Guide provides more details on each stall reason.

Instruction Statistics

Statistics of the executed low-level assembly instructions (SASS). The instruction mix provides insight into the types and frequency of the executed instructions. A narrow mix of instruction types implies a dependency on few instruction pipelines, while others remain unused. Using multiple pipelines allows hiding latencies and enables parallel execution. Note that 'Instructions/Opcode' and 'Executed Instructions' are measured differently and can diverge if cycles are spent in system calls.

Executed Instructions [inst]	524288	Avg. Executed Instructions Per Scheduler [inst]	1024
Issued Instructions [inst]	553335	Avg. Issued Instructions Per Scheduler [inst]	1080.73

NVLink Topology

NVLink Topology diagram shows logical NVLink connections with transmit/receive throughput.

NVLink Tables

Detailed tables with properties for each NVLink.

Metric Details

Pin Tab | Search metrics in current report

Open an Nsight Compute profiling report, then click a metric on the Details or Raw page.
Use the search bar to lookup metrics in the current report.

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NVIDIA Nsight Compute

File Connection Debug Profile Tools Window Help

Connect | Disconnect | Terminate | Profile Kernel | Baselines | Metric Details

Project Explorer

Search project...

Default Project

Welcome | test.ncu-rep | Untitled 1 *

Page: Details | Result: 3 - 559 - transpose_naive | Add Baseline | Apply Rules | Occupancy Calculator | Copy as Image

	Result	Time	Cycles	Regs	GPU	SM Frequency	CC	Process
Current	559 - transpose_naive (64, 64, 1)x(8, 32, 1)	8.80 usecond	19,529	16	0 - NVIDIA GeForce RTX 4090	2.21 cycle/nsecond	8.9	[544] my_transpose

Issued Instructions [inst]	553335	Avg. Issued Instructions Per Scheduler [inst]	1080.73

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NVLink Topology diagram shows logical NVLink connections with transmit/receive throughput.

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NUMA Affinity

Non-uniform memory access (NUMA) affinities based on compute and memory distances for all GPUs.

Launch Statistics

Summary of the configuration used to launch the kernel. The launch configuration defines the size of the kernel grid, the division of the grid into blocks, and the GPU resources needed to execute the kernel. Choosing an efficient launch configuration maximizes device utilization.

Grid Size	4096	Function Cache Configuration	CachePreferNone
Registers Per Thread [register/thread]	16	Static Shared Memory Per Block [byte/block]	0
Block Size	256	Dynamic Shared Memory Per Block [byte/block]	0
Threads [thread]	1048576	Driver Shared Memory Per Block [Kbyte/block]	1.02
Waves Per SM	5.33	Shared Memory Configuration Size [Kbyte]	16.38

Occupancy

Occupancy is the ratio of the number of active warps per multiprocessor to the maximum number of possible active warps. Another way to view occupancy is the percentage of the hardware's ability to process warps that is actively in use. Higher occupancy does not always result in higher performance, however, low occupancy always reduces the ability to hide latencies, resulting in overall performance degradation. Large discrepancies between the theoretical and the achieved occupancy during execution typically indicates highly imbalanced workloads.

Theoretical Occupancy [%]	100	Block Limit Registers [block]	16
Theoretical Active Warps per SM [warp]	48	Block Limit Shared Mem [block]	16
Achieved Occupancy [%]	72.94	Block Limit Warps [block]	6
Achieved Active Warps Per SM [warp]	35.01	Block Limit SM [block]	24

Occupancy Limiters
Est. Speedup: 27.06%

This kernel's theoretical occupancy is not impacted by any block limit. The difference between calculated theoretical (100.0%) and measured achieved occupancy (72.9%) can be the result of warp scheduling overheads or workload imbalances during the kernel execution. Load imbalances can occur between warps within a block as well as across blocks of the same kernel. See the CUDA Best Practices Guide for more details on optimizing occupancy.

Source Counters

Source metrics, including branch efficiency and sampled warp stall reasons. Warp Stall Sampling metrics are periodically sampled over the kernel runtime. They indicate when warps were stalled and couldn't be scheduled. See the documentation for a description of all stall reasons. Only focus on stalls if the schedulers fail to issue every cycle.

Branch Instructions [inst]	32768	Branch Efficiency [%]	0
Branch Instructions Ratio [%]	0.06	Avg. Divergent Branches	0

Uncoalesced Global Accesses

This kernel has uncoalesced global accesses resulting in a total of 131072 excessive sectors (33% of the total 393216 sectors). Check the L2 Theoretical Sectors Global Excessive table for the primary source locations. The CUDA Programming Guide had additional information on reducing uncoalesced device memory accesses.

Metric Details

Pin Tab | Search metrics in current report

sm__warps_active.avg.pct_of_peak_sustained_active

Name	sm__warps_active.avg.pct_of_peak...
Unit	%
Value	72.9355309809528
Report	test.ncu-rep

Additional Information

Description:

cumulative # of warps in flight
(This counter metric represents the avg percent of the peak sustained rate achieved during unit active cycles across all sub-unit instances)

Knowledgebase Entry:

sm: The Streaming Multiprocessor handles execution of a kernel as groups of 32 threads, called warps.
Warps are further grouped into cooperative thread arrays (CTA), called blocks in CUDA.

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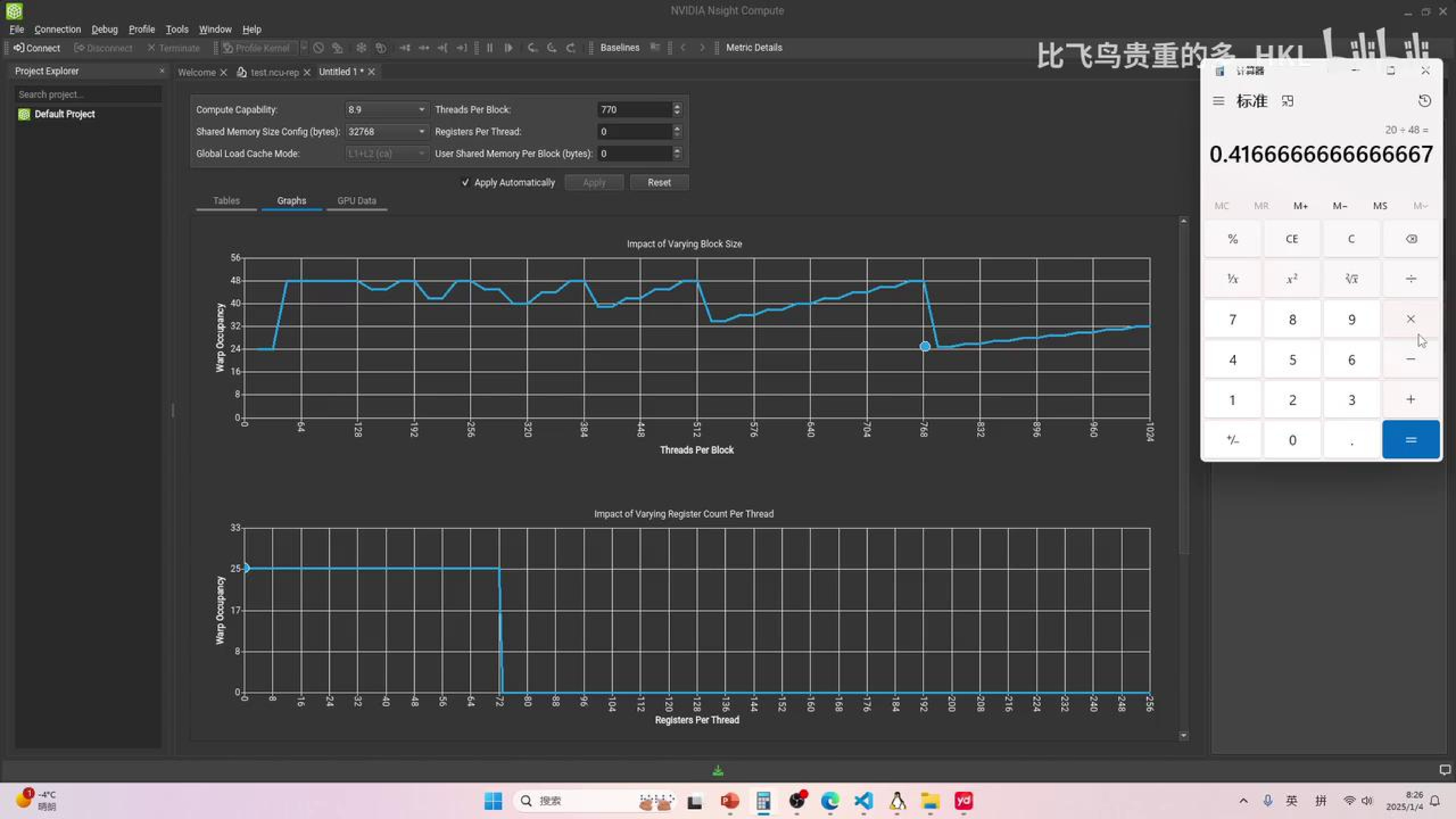
NVIDIA Nsight Compute
File Connection Debug Profile Tools Window Help
Connect Disconnect Terminate Profile Kernel Baselines Metric Details
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Project Explorer
Search project...
Default Project
Welcome test.ncu-rep Untitled 1 *
Compute Capability: 8.9
Threads Per Block: 770
Shared Memory Size Config (bytes): 32768
Registers Per Thread: 0
Global Load Cache Mode: L1+L2 (ca)
User Shared Memory Per Block (bytes): 0
Apply Automatically Apply Reset
Tables Graphs GPU Data
Impact of Varying Block Size
Warp Occupancy
Threads Per Block
Impact of Varying Register Count Per Thread
Warp Occupancy
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Add Baseline Apply Rules Occupancy Calculator Copy as Image
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Impact of Varying Register Count Per Thread
Warp Occupancy
Registers Per Thread
Impact of Varying Block Size
Warp Occupancy
Block Size
Impact of Varying Shared Memory Usage Per Block
Warp Occupancy
Shared Memory Per Block
Source Counters
Source metrics, including branch efficiency and sampled warp stall reasons. Warp Stall Sampling metrics are periodically sampled over the kernel runtime. They indicate when warps were stalled and couldn't be scheduled. See the documentation
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Pin Tab Search metrics in current report
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Name sm__warps_active.avg.pct_of_peak...
Unit %
Value 72.9355309809528
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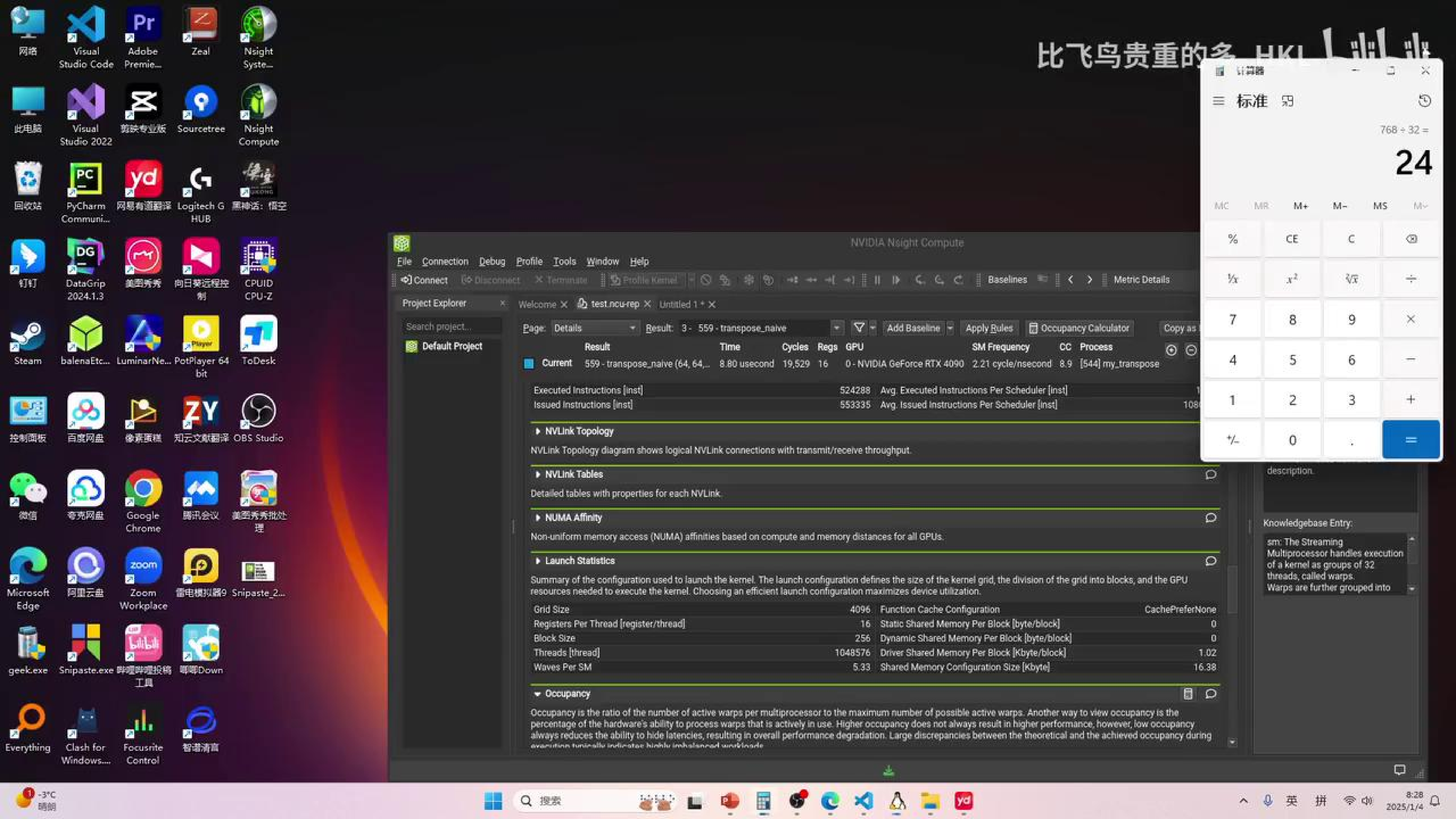

NVIDIA Nsight Compute
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Registers Per Thread [register/thread] 16 Static Shared Memory Per Block [byte/block] 0
Block Size 256 Dynamic Shared Memory Per Block [byte/block] 0
Threads [thread] 1048576 Driver Shared Memory Per Block [Kbyte/block] 1.02
Waves Per SM 5.33 Shared Memory Configuration Size [Kbyte] 16.38
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Occupancy is the ratio of the number of active warps per multiprocessor to the maximum number of possible active warps. Another way to view occupancy is the percentage of the hardware's ability to process warps that is actively in use. Higher occupancy does not always result in higher performance, however, low occupancy always reduces the ability to hide latencies, resulting in overall performance degradation. Large discrepancies between the theoretical and the achieved occupancy during
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计算器
标准
768 ÷ 32 =
24
MC MR M+ M- MS
% CE C
7 8 9
4 5 6
1 2 3
+/- 0 . =